Plugging into the Future: An Exploration of Electricity Consumption Patterns using Tableau

INTRODUCTION

a) Project Overview:

Plugging into the future:An exploration of electricity consumption patterns project focuses on analyzing electricity consumption patterns using Data Analytics techniques and visualizing insights through Tableau. Electricity is one of the most important resources in modern society, and understanding its usage patterns helps in improving efficiency, reducing wastage, and planning future energy needs.

The main objective of this project is to study electricity consumption data and identify trends, seasonal variations, peak usage hours, and consumption behavior across different regions or sectors. By applying data analytics techniques, raw electricity data is transformed into meaningful insights.

b) Purpose:

The purpose of the project “Plugging into the Future: An Exploration of Electricity Consumption Patterns” is to analyze and understand how electricity is consumed across different regions, seasons, and time periods, and to convert this data into meaningful insights. The project aims to identify trends such as peak usage hours, monthly and yearly growth patterns, and regional differences in electricity demand. By using data visualization tools, it transforms complex raw data into clear and interactive dashboards that are easy to interpret. This helps households, energy analysts, and government authorities make informed decisions, reduce electricity costs, improve energy efficiency, and plan future power supply effectively. Overall, the project supports smarter energy management and promotes sustainable electricity usage for the future.

IDEATION PHASE

a) Define the Problem Statements:

Date	10 February 2026

Team ID	LTVIP2026TMIDS36768
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Marks	2 Marks

Customer Problem Statements:

Problem Statement (PS-1):

I am a State Electricity Board Officer trying to identify how electricity consumption varies across states, regions, and time periods. However, I lack a visual and interactive dashboard system to effectively analyze the data. The raw electricity dataset is large, complex, and difficult to interpret using traditional tools like Excel. Because of this, I am often unsure about peak demand periods and face inefficiencies in planning and distributing power resources, which makes me feel uncertain and less confident in making accurate, data-driven decisions.

Problem Statement (PS-2):

I am an Energy Policy Analyst trying to evaluate seasonal trends, regional usage patterns, and yearly growth in electricity consumption. However, I do not have a centralized system to visually compare multiple dimensions such as state, region, month, and quarter. There is also a lack of dynamic, real-time visualization tools that allow interactive filtering and comparison of data. Because of this, I often feel underprepared to make accurate, data-driven energy policy decisions and effective infrastructure planning.

Example:

Problem	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a State Electricity Board Officer	identify how electricity consumption varies across states, regions, and time periods	I lack a visual and interactive dashboard system	the raw electricity dataset is large, complex, and difficult to analyze using traditional tools like Excel	unsure about peak demand periods and inefficient in planning power distribution
PS-2	an Energy Policy Analyst	evaluate seasonal trends, regional usage patterns, and yearly growth in electricity consumption	. I don't have a centralized way to visually compare multiple dimensions like state, region, month, and quarter	there is a lack of dynamic, real-time visualization tools that allow interactive filtering and comparison	underprepared to make data-driven energy policy decisions and infrastructure plannin

b) Empathy Map Canvas:

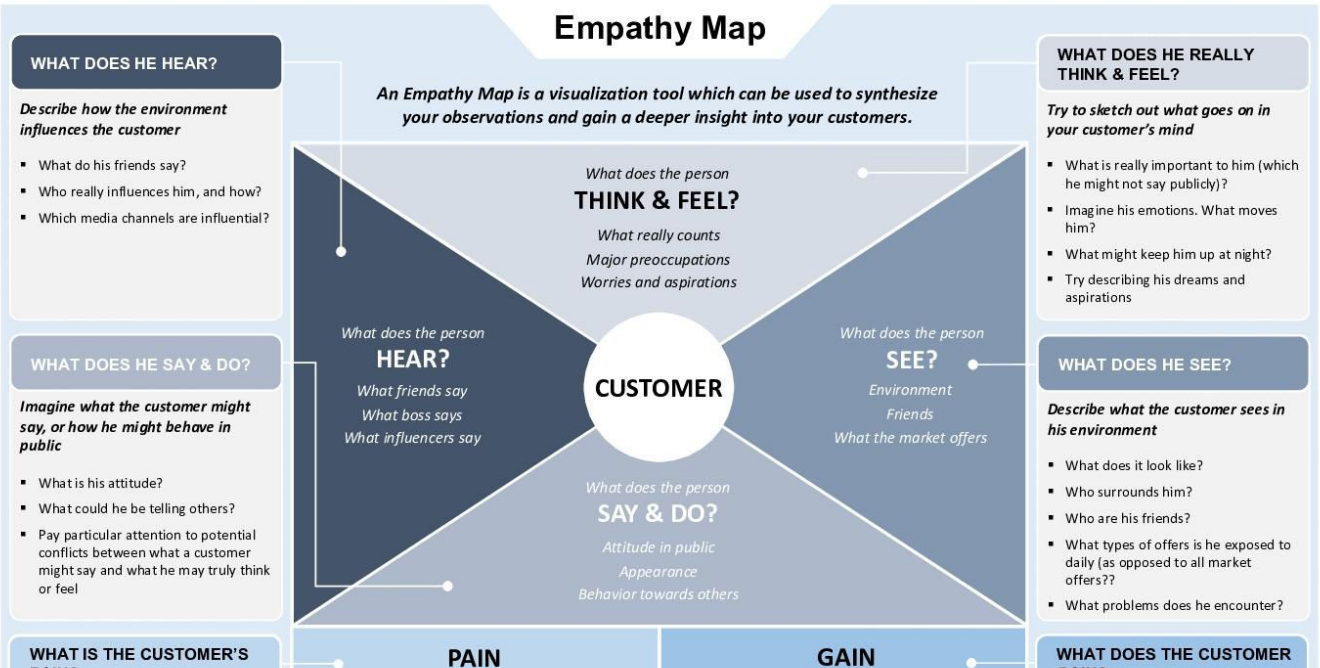
Date	10 February 2026
Team ID	LTVIP2026TMIDS36768

Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Maximum Marks	4 Marks

Empathy Map Canvas:

The empathy map for Plugging into the future: An exploration of electricity consumption patterns project focuses on electricity consumers such as households and small business owners who are directly affected by changing energy consumption patterns. These users see their monthly electricity bills, smart meter readings, and increasing usage of electrical appliances in daily life. They hear news about rising electricity tariffs, energy conservation campaigns, and advice from others about reducing power consumption. They think and feel concerned about high electricity costs, confused about sudden bill increases, and motivated to manage their usage more efficiently. They often say things like, “My electricity bill is too high,” and take actions such as turning off unused appliances, using energy-efficient devices, or monitoring their consumption more carefully. However, they experience pains like lack of clear understanding of usage trends, unexpected spikes in bills, and difficulty identifying peak consumption periods. Their gains include wanting accurate insights, clear visual reports, cost-saving suggestions, and better control over electricity usage. This project addresses needs by analyzing electricity consumption data and presenting meaningful visualizations, helping users make informed and sustainable energy decisions.

Example:



Example:

Plugging into the future:An exploration of electricity consumption patterns

c) Brainstorming:

Date	12 February 2026
Team ID	LTVIP2026TMIDS36768
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization Template:

Problem Statement Chosen:

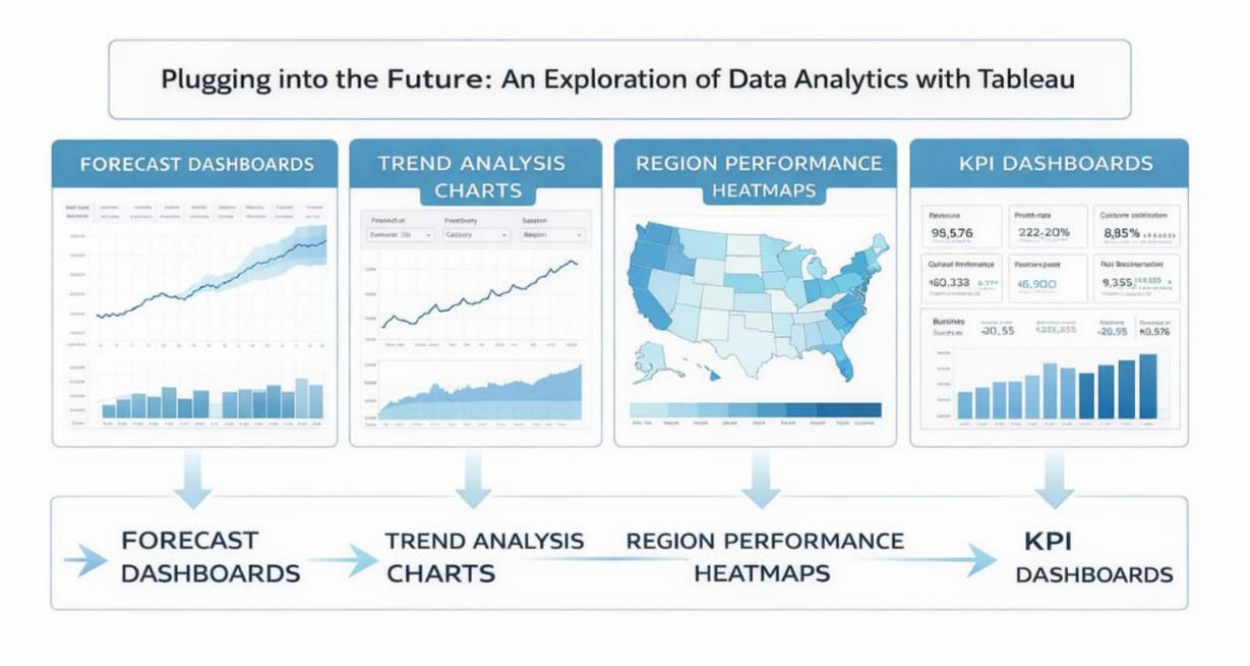
Government energy departments and electricity boards struggle to clearly analyze electricity

Consumption patterns across states and regions due to scattered data and lack of interactive visualization

Our team came together to understand the electricity consumption dataset containing state-wise, region-

Wise, monthly, quarterly, and yearly usage data for 2019 and 2020.

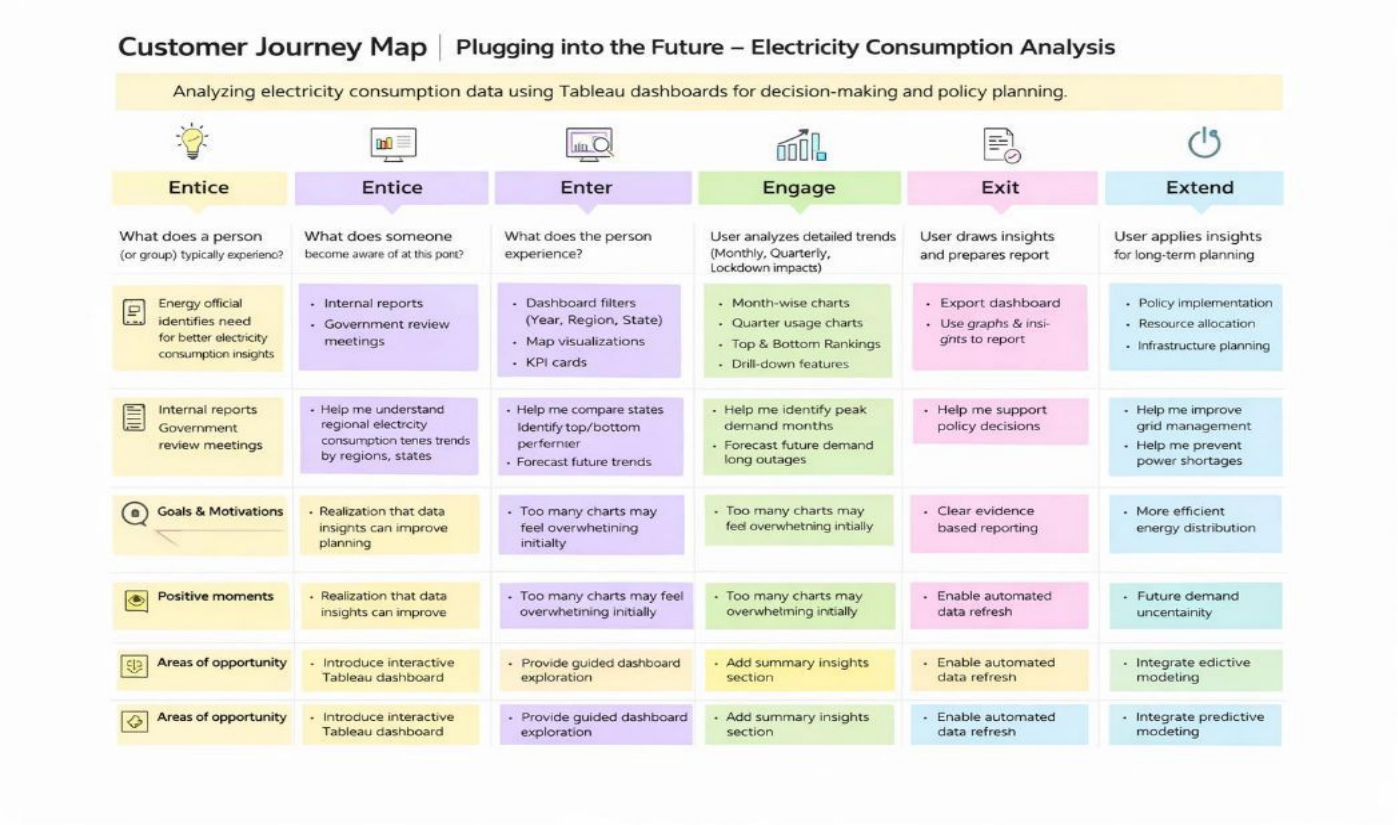
Step-1: Team Gathering, Collaboration and Select the Problem Statement



REQUIREMENT ANALYSIS

a)Customer Journey Map

Plugging into the future:An exploration of electricity consumption patterns



b)Solution Requirements (Functional & Non-functional):

Date	12 February 2026
Team ID	LTVIP2026TMIDS36768
Project Name	plugging into the future : an explanation of electricity consumption using tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR- no.	Functional Requirement	Sub - Requirement (Story / Sub -task)
FR - 1	Data collection and Extraction from Database	Collect the Dataset Storing Data in DB and perform SQL Operations Connect DB with Tableau
FR - 2	Data Visualization	No of Unique Visualizations
FR - 3	Dashboard	No of Unique Visualizations
FR - 4	Story	Responsive and Design of Dashboard
FR -5	Performance Testing	Amount of Data Rendered to DB Utilization of Data Filters No of Visualizations / Graphs
FR - 6	Web Integration	Dashboard and Story Embed with UI with Flask

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR - no.	Non - Functional Requirement	Description
NFR - 1	Usability	The dashboard should be simple, user-friendly, and easy to understand.
NFR - 2	Security	Electricity data must be securely stored and protected from unauthorized access.

NFR - 3	Reliability	The system should provide accurate and consistent results.
NFR - 4	Performance	The dashboard should load quickly and respond smoothly.
NFR - 5	Availability	The system should be accessible whenever users need it.
NFR - 6	Scalability	The system should handle increasing electricity data in the future without performance issues.

c)Data Flow Diagram :

Date	13 February 2026
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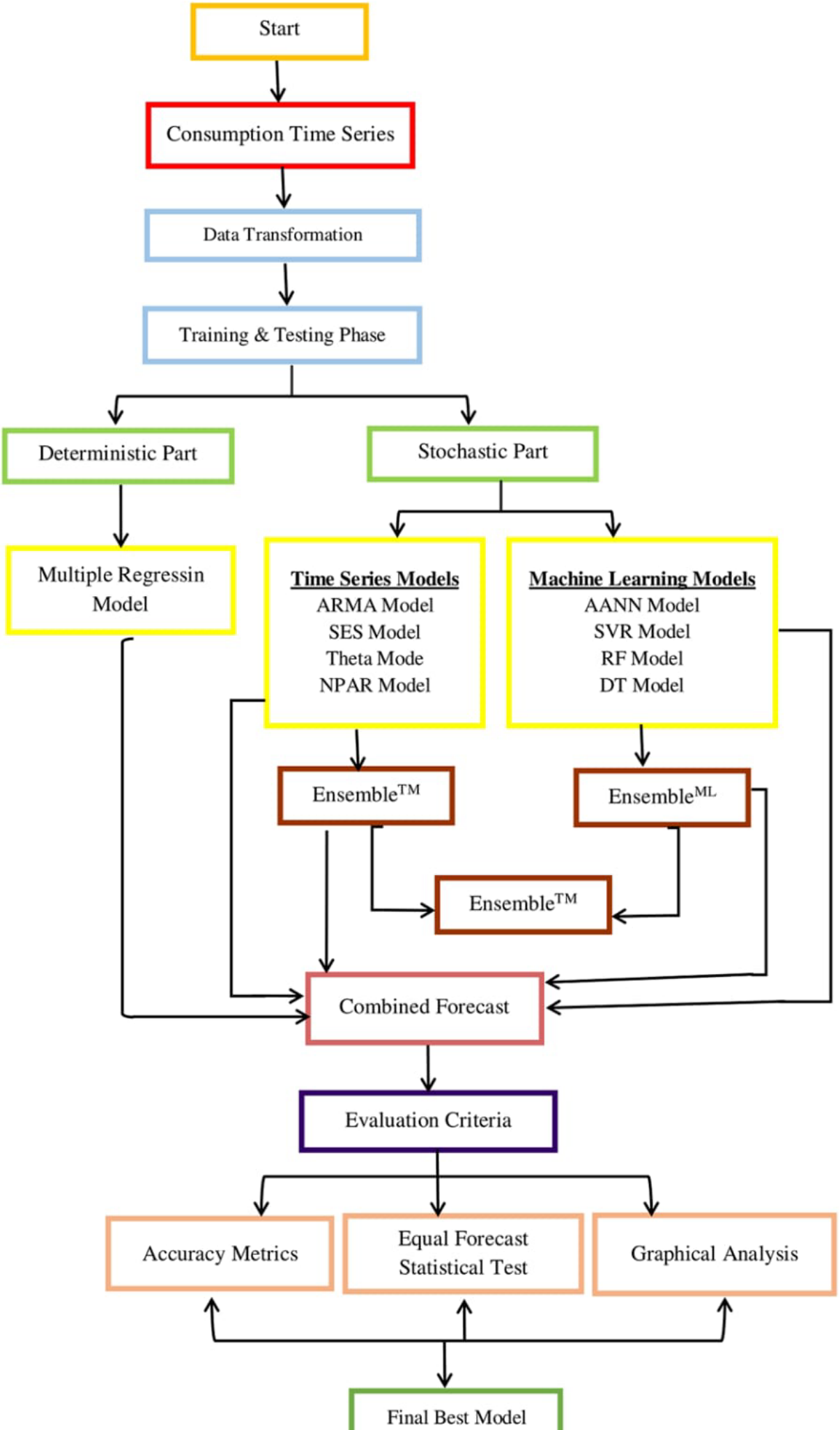
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) outlines how raw energy data is collected, processed, and transformed into actionable insights in tableau. It emphasizes the journey from traditional energy consumption to future forecasting and renewable integration.


Start
Consumption Time Series
Data Transformation
Training & Testing Phase
Deterministic Part
Stochastic Part
Multiple Regressin Model
Time Series Models
ARMA Model
SES Model
Theta Mode
NPAR Model
Machine Learning Models
AANN Model
SVR Model
RF Model
DT Model
EnsembleTM
EnsembleML
EnsembleTM
Combined Forecast
Evaluation Criteria
Accuracy Metrics
Equal Forecast Statistical Test
Graphical Analysis
Final Best Model

d) Technology Stack (Architecture & Stack)

Date	12 February 2026
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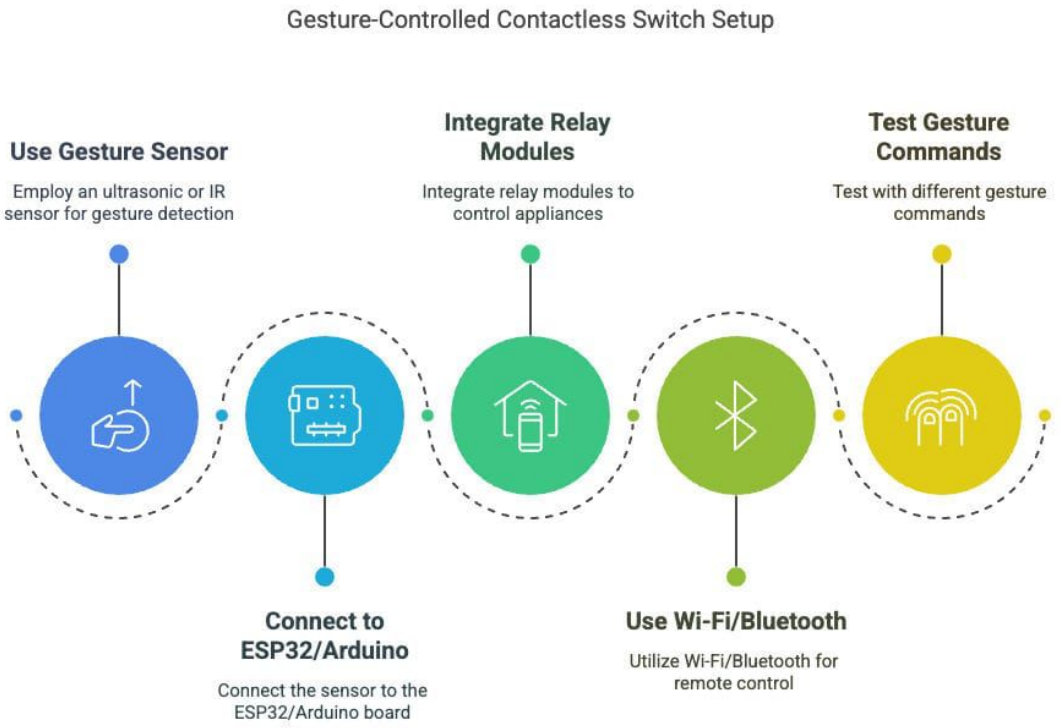
Maximum Marks	4 Marks

Technical architecture

This technical architecture outlines a modern data stack designed to explore, analyze, and forecast future electricity consumption using Tableau. The project focuses on integrating high-velocity sensor data with historical datasets to provide interactive insights.



PROJECT DESIGN

a) Problem – Solution Fit Template:

Date	13 February 2026
Project id	LTVIP2026TMIDS36768
Project name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using Tableau
Minimum marks	**2 marks**

Problem – Solution Fit :

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem.
It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- Solve complex problems in a way that fits the state of your customers.
- Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- Sharpen your communication and marketing strategy with the right triggers and messaging.

- Increase touch-points with your company by finding the right problem-behavior fit and
- building trust by solving frequent annoyances, or urgent or costly problems.
- Understand the existing situation in order to improve it for your target roup.

Template:

b)Proposed Solution Template :

Date	12 February 2026
Project id	LTVIP2026TMIDS36768
Project name	Plugging into the Future: An Exploration of Electricity Consumption Patterns Using

	Tableau
Maximum marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S. No	Parameter	Description
1	Problem statement (Problem to be solved)	Electricity consumption is increasing rapidly, and there is a lack of clear analysis to understand usage patterns, peak demand hours, and energy wastage. This leads to higher costs and grid instability.
2	Idea / Solution description	The project uses Tableau to analyze and visualize electricity consumption data through interactive dashboards, identifying trends, peak usage times, and seasonal variations. Using Tableau dashboards, smart meter data is transformed into clear visual insights about electricity usage patterns.
3	Novelty / Uniqueness	Real-time interactive visualization with easy-to-understand graphs helps decision-makers quickly identify consumption trends and take action.
4	Social impact / customer satisfaction	Improves energy awareness among consumers, reduces electricity bills, and ensures better power supply management.
5	Business model	Utility companies can use this system as a

	(Revenue model)	subscription-based analytics service to improve operational efficiency and reduce energy losses.
6	Scalability of the solution	The solution can be scaled from small residential areas to smart cities by integrating more smart meters and expanding dashboard analytics.

c)Solution Architecture:

Date	13 February 2026
Project id	LTVIP2026TMIDS36768
Project name	Plugging into the future :An exploration of data analyticals with tableau
Minimum marks	**4 marks**

Solution Architecture:

<u>1.Data Sources Layer:</u>

Collect electricity consumption data from:

- Smart meters (household usage)
- Industrial meters
- Weather data APIs
- Historical electricity records
- Grid load data

Data Types:

- Timestamp
- Units consumed (kWh)
- Location

- Consumer type
- Temperature
- Peak/Off-peak indicators

2.Data Ingestion Layer

This layer collects and transfers data into the system.

Tools Used:

- API integration
- IoT gateways
- Batch uploads (CSV/Excel)
- Streaming ingestion tools

Real-time ingestion tools:

- Apache Kafka
- Apache NiFi

Purpose:

- Ensure continuous data flow
- Handle real-time smart meter data

3.Data Storage Layer

Stores both raw and processed data.

Storage Options:

- Cloud Data Lake
- Data Warehouse
- Example platforms:
- Amazon S3
- Google BigQuery
- Microsoft Azure SQL Database

Data Storage Types:

- Raw Data (Unprocessed)
- Cleaned Data
- Aggregated Data

4.Data Processing & Analytics Layer

This is the core intelligence layer.

Data Cleaning:

- Remove missing values

- Handle anomalies
- Normalize data

Data Transformation:

- Daily/Weekly/Monthly aggregation
- Peak hour detection
- Consumer segmentation

Advanced Analytics:

- Trend Analysis
- Load Forecasting
- Seasonal Pattern Detection
- Demand Prediction using ML models

Tools:

- Apache Spark
- Python
- TensorFlow

5. Visualization & Reporting Layer

Transforms insights into interactive dashboards.

Tools:

- Power BI
- Tableau

Dashboards show:

- Daily usage trends
- Monthly growth patterns
- Peak load analysis
- Regional consumption comparison
- Future demand prediction

example:

Vijayawada Smart City Electricity Monitoring

- 10,000 homes have smart meters
- Data collected every 15 minutes

System finds:

- High AC usage in summer
- Peak load at night

Insight:

Government introduces:

- Time-based electricity pricing
- Energy-saving awareness programs

Result:

- 15% reduction in peak load
- Better power grid stability

Example:Solution Architecture

Plugging into the Future:
An Exploration of Electricity Consumption Patterns
Energy Sources
Renewables
Fossil Fuels
Nucear
Hydroelectric
Consumption by Sector
Residential 30%
Industrial 25%
Industrial 35%
Industrial 35%
Usage Trends
Energy Efficiency
Smart Meters
Home Automation
Efficient Appliances
Environmental Impact
CO2 Emissions
Renewable Adoption
Sustainability Goals

PROJECT PLANNING AND SCHEDULING

Planning Logic for " Plugging into the future:An exploration of electricity consumption patterns using Tableau project"

A Sprint is a fixed period or duration in which a team works to complete a set of tasks.
An Epic is a large objective (e.g., full Tableau dashboard) that is broken into smaller, manageable Stories.
Story Points represent the relative effort needed to complete a story, often using Fibonacci sequence (1, 2, 3,
5...).

Sprint 1 (5 Days):
Epic - Data Preparation & Understanding:

Task	Description	Story points
Data collection	Collect smart meter data, household usage, seasonal data (CSV, govt portals)	2
Data cleaning	Handle missing values, remove duplicates, correct units (kWh)	3
Data Transformation	Create calculated fields (daily avg, peak load, monthly usage)	2
Exploratory Data Analysis	Identify peak hours, seasonal trends, high consumption areas	2

Sprint 1 Total Story Points = 9

Sprint 2 (5 Days)

Epic: Visualization Development:

Task	Description	Story points
Dashboard Design	Plan layout (KPIs, bar charts, line graphs, heat maps)	**3**
Chart Development	Build visuals in Tableau (Consumption trends, Peak usage, Region comparison)	**5**
Interactivity Integration	Add filters (date, region, household type), tooltips, drill-down	**3**
Testing & Refinement	Check data accuracy and dashboard responsiveness	**2**

Sprint 2 Total Story Points = 13

Sprint 3 (5 Days)

Epic: Insight Presentation & Deployment

Task	Description	Story points
Storyboard Creation	Add insights like "Peak usage at 8 PM", "Summer AC impact"	2
User Feedback Review	Collect suggestions from stakeholders	3
Final Revisions	Improve visuals, clarity, performance	3
Public Deployment	Publish dashboard online	2

Sprint 3 Total Story Points = 10

Summary & Velocity Calculation:

- Total Story Points = 9 (Sprint 1) + 13 (Sprint 2) + 10 (Sprint 3)
- Total = 32 Story Points
- Total Sprints = 3
- Average Velocity = 32 / 3 = 10–11 Story Points per Sprint

FUNCTIONAL AND PERFORMANCE TESTING

Model Performance Test

Date	15 February 2026
Project id	LTVIP2026TMIDS36768
Project name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum marks	

Model Performance Testing:

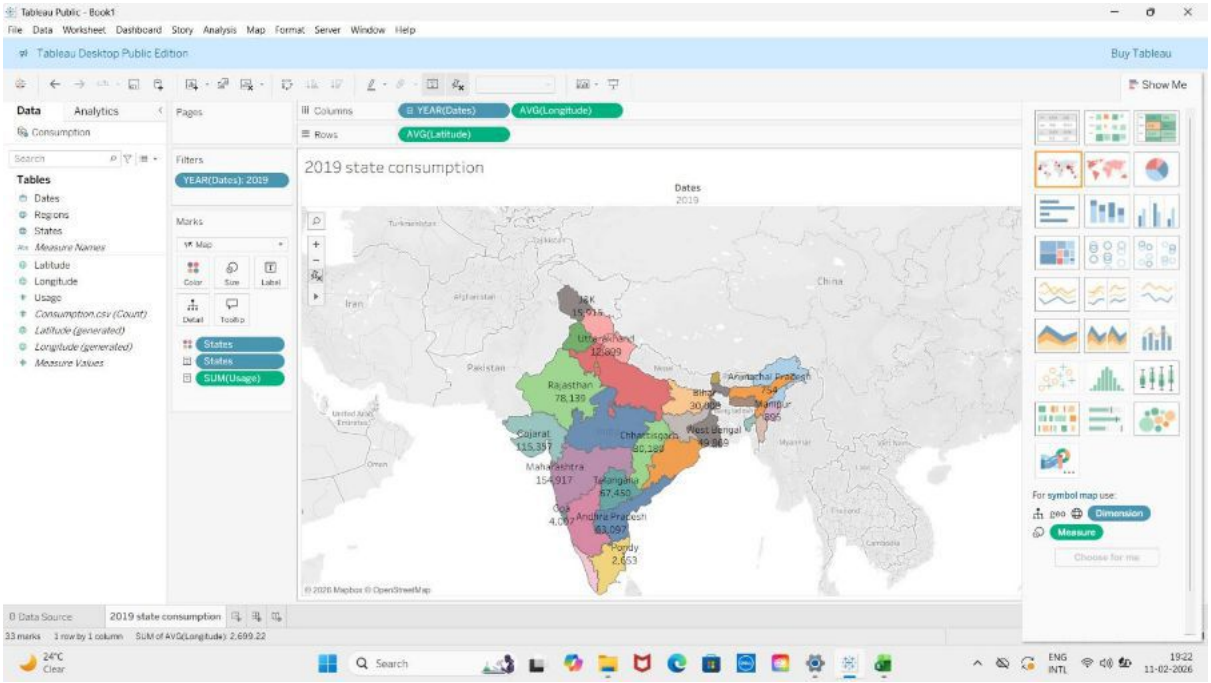
Project team shall fill the following information in model performance testing template

S.No	Parameter	Screen shot / values
1	Data Rendered	Displayed electricity consumption data across regions, households, and time periods. Shows rendered line graph (monthly trend), bar chart (region-wise usage), heat map (peak hours), and KPI cards (Total kWh, Avg Usage, Peak Load).
2	Data Preprocessing	Removed null values, standardized date formats, converted units to kWh, handled outliers, created date

		hierarchy (Year > Quarter > Month > Day).
3	Utilization of Filters	Filters applied: Region, Year, Month, Consumer Type (Residential/Commercial), Tariff Type. Enables users to analyze peak usage, seasonal trends, and area-wise consumption.
4	Calculation Fields Used	Peak Load = MAX(kWh) Average Consumption = SUM(kWh)/COUNT(Days) Consumption Growth (%) = (Current - Previous) / Previous Load Factor = Average Load / Peak Load
5	Dashboard Design	No. of Visualizations / Graphs – 5 (Line Chart – Trend Analysis, Bar Chart – Region Comparison, Heat Map – Hourly Usage, Pie Chart – Consumer Share, KPI Cards – Summary Metrics).
6	Story Design	Consumption Overview, Regional Analysis, Seasonal Demand Pattern, Peak Hour Analysis, Energy Saving Insights).

RESULTS

Screenshots of Visualizations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SUM(Usage)
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2019 state consumption
Dates
2019
China
Pakistan
Rajasthan
78,139
Gujarat
115,357
Maharashtra
154,917
Telangana
67,450
Andhra Pradesh
63,097
Pondy
2,653
Arunachal Pradesh
Manipur
West Bengal
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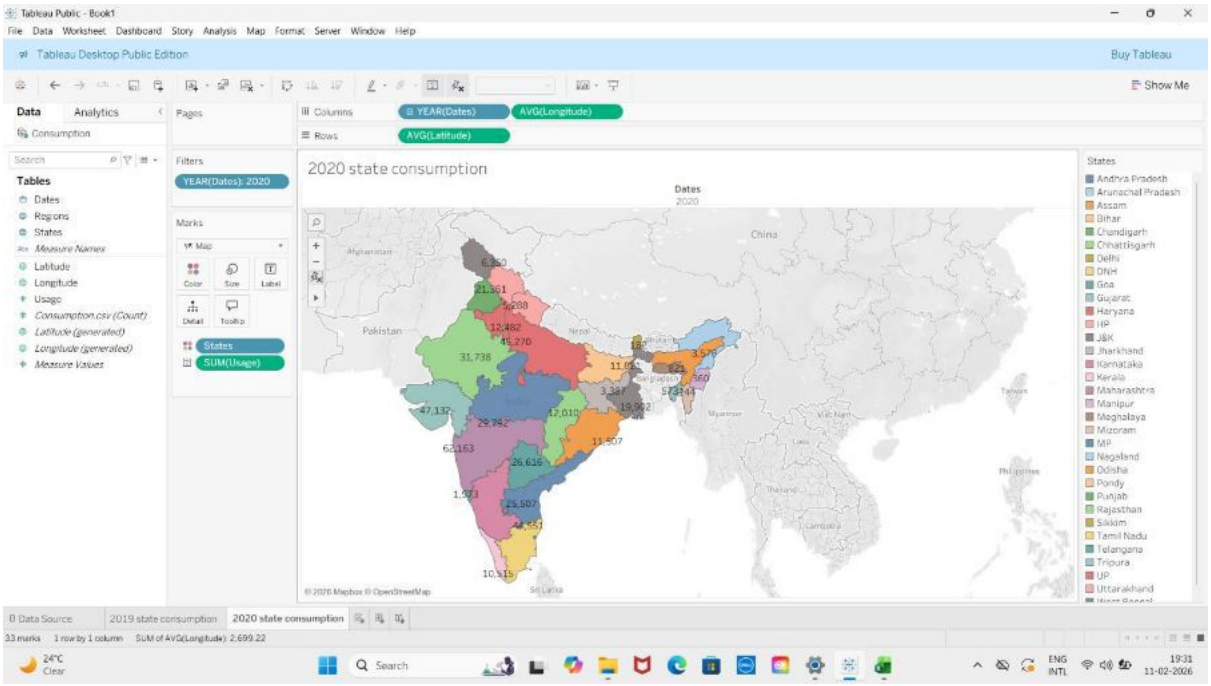
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2020 state consumption
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2020
China
Pakistan
31,738
47,132
62,163
29,942
12,010
19,902
11,507
26,616
25,507
10,515
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Andhra Pradesh
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Assam
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Chandigarh
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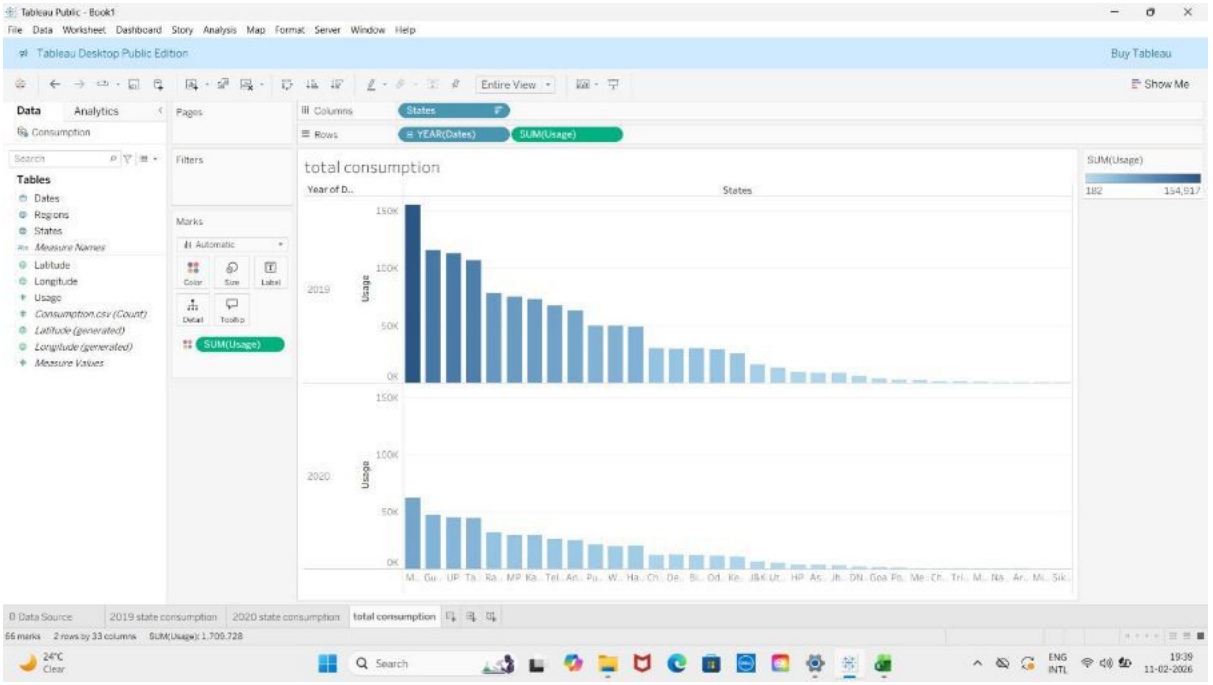
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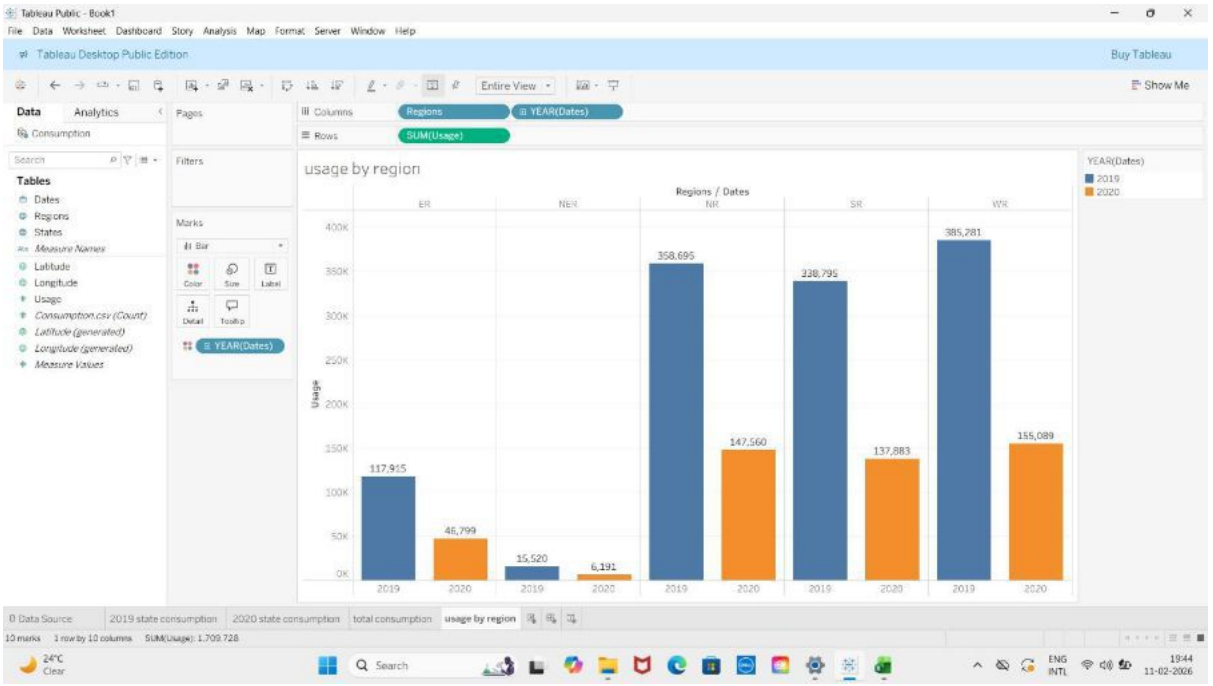
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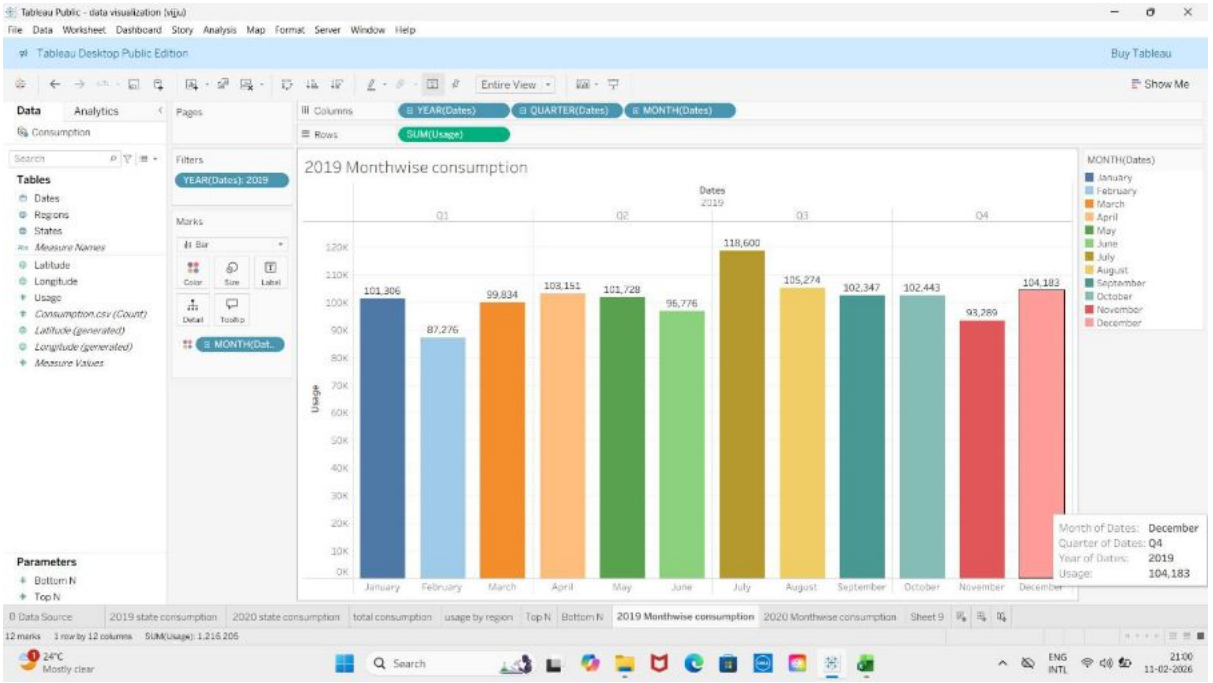
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90K
80K
70K
60K
50K
40K
30K
20K
10K
0K
101,306
87,276
99,834
103,151
101,728
96,776
118,600
105,274
102,347
102,443
93,289
104,183
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December
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Quarter of Dates: Q4
Year of Dates: 2019
Usage: 104,183
0 Data Source
2019 state consumption
2020 state consumption
total consumption
usage by region
Top N
Bottom N
2019 Monthwise consumption
2020 Monthwise consumption
Sheet 9
12 marks 1 row by 12 columns SUM(Usage): 1,216,205
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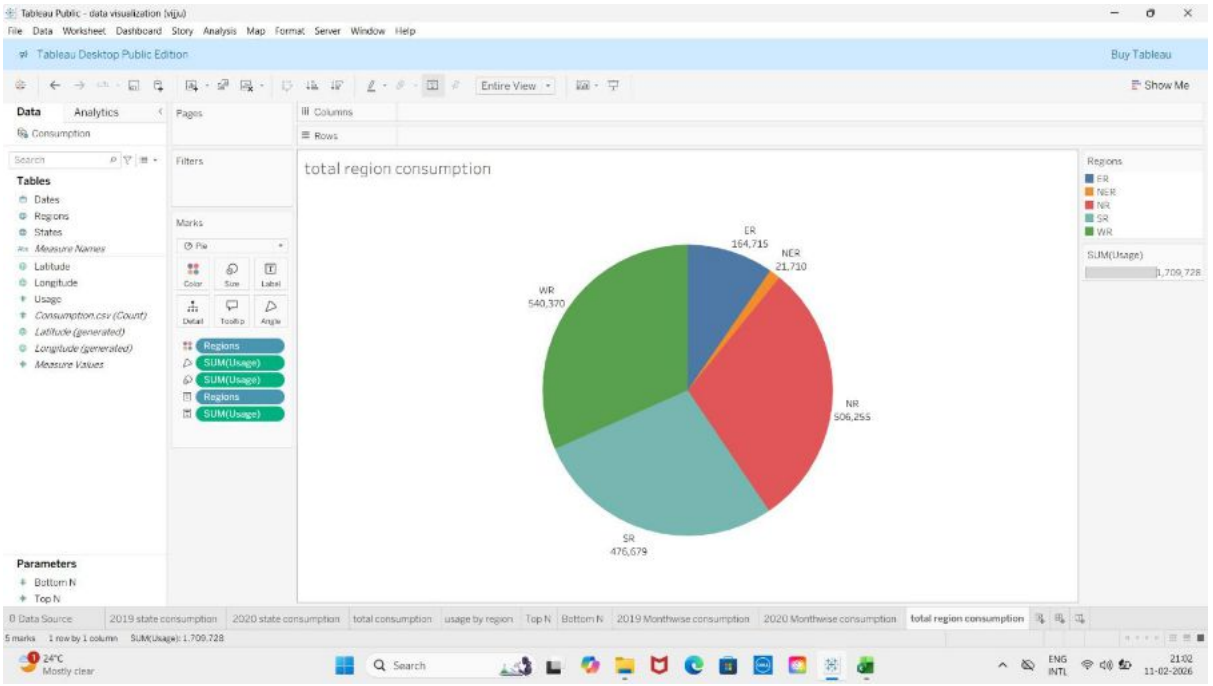
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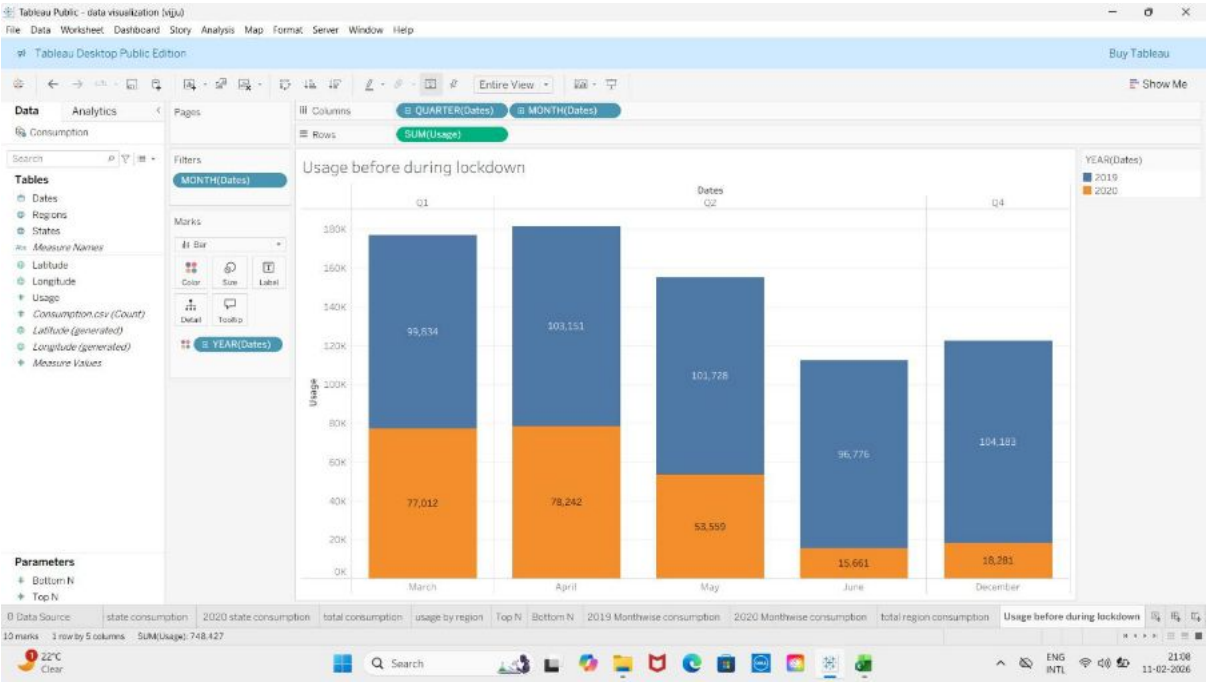
Data
Analytics
Consumption
Tables
Dates
Regions
States
Measure Names
Latitude
Longitude
Usage
Consumption.csv (Count)
Latitude (generated)
Longitude (generated)
Measure Values
Parameters
Bottom N
Top N
Pages
Filters
Marks
Pie
Color
Size
Label
Detail
Tooltip
Angle
Regions
SUM(Usage)
SUM(Usage)
Regions
SUM(Usage)
Columns
Rows
total region consumption
ER
164,715
NER
21,710
NR
506,255
SR
476,679
WR
540,370
Regions
ER
NER
NR
SR
WR
SUM(Usage)
1,709,728
0 Data Source
2019 state consumption
2020 state consumption
total consumption
usage by region
Top N
Bottom N
2019 Monthwise consumption
2020 Monthwise consumption
total region consumption
5 marks 1 row by 1 column SUM(Usage): 1,709,728
24°C
Mostly clear
Search
ENG
INTL
21:02
11-02-2026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Consumption.csv (Count)
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Measure Values
Parameters
Bottom N
Top N
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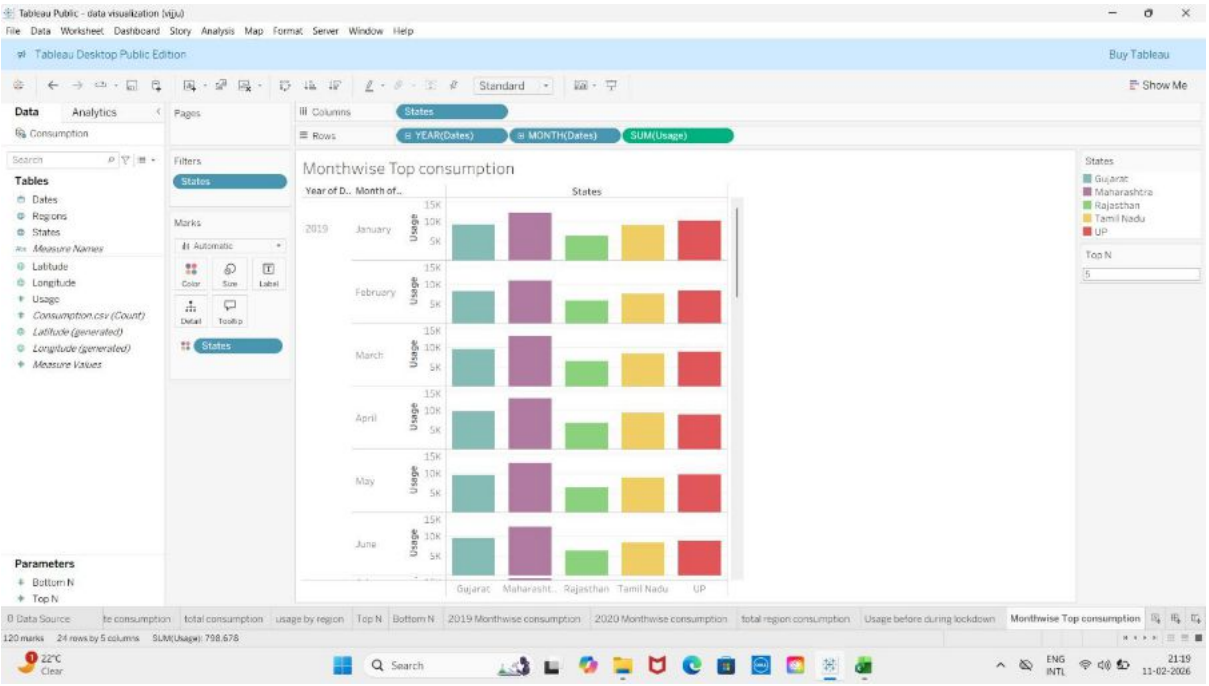
Filters
MONTH(Dates)
Marks
Bar
Color
Size
Label
Detail
Tooltip
YEAR(Dates)
Columns
QUARTER(Dates)
MONTH(Dates)
Rows
SUM(Usage)
Usage before during lockdown
Dates
Q1
Q2
Q4
Usage
180K
160K
140K
120K
100K
80K
60K
40K
20K
0K
99,634
77,012
103,151
78,242
101,728
53,559
96,776
15,661
104,183
18,281
March
April
May
June
December
YEAR(Dates)
2019
2020
Data Source
state consumption
2020 state consumption
total consumption
usage by region
Top N
Bottom N
2019 Monthwise consumption
2020 Monthwise consumption
total region consumption
Usage before during lockdown
10 marks
1 row by 5 columns
SUM(Usage): 748,427
22°C
Clear
Search
ENG
INTL
21:08
11-02-2026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Measure Values
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Color
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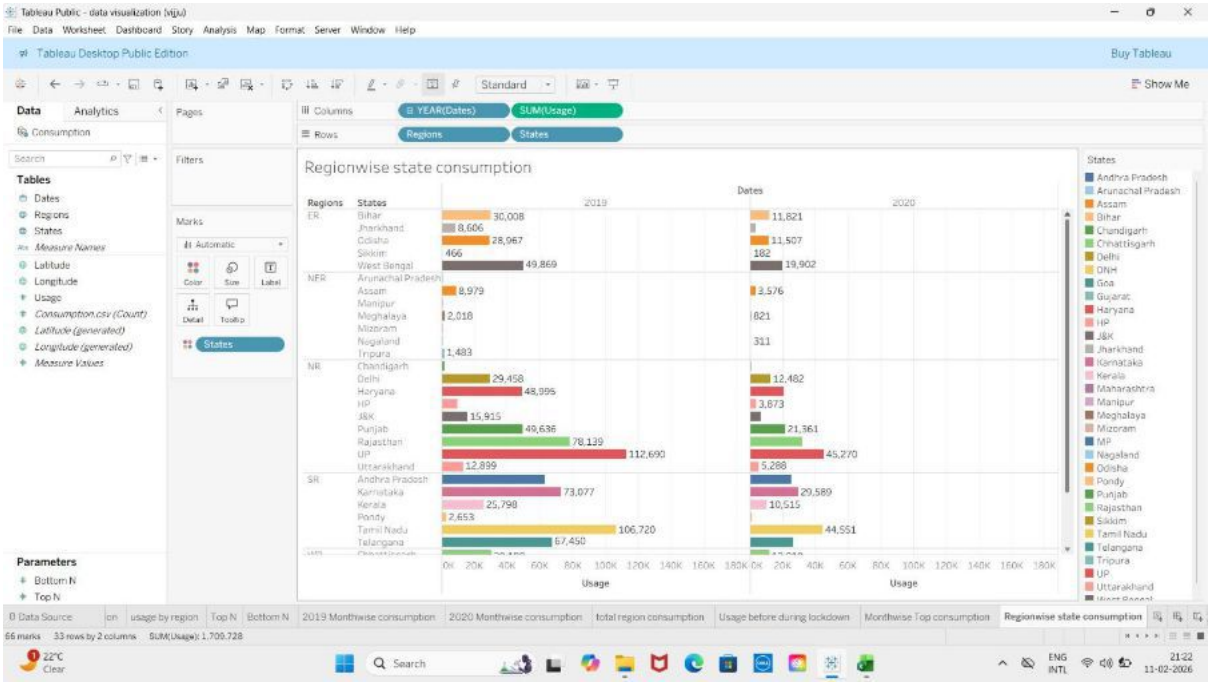
Label
Detail
Tooltip
States
Columns
States
Rows
YEAR(Dates)
MONTH(Dates)
SUM(Usage)
Monthwise Top consumption
Year of D..
Month of..
States
2019
January
February
March
April
May
June
Usage
15K
10K
5K
Gujarat
Maharasht..
Rajasthan
Tamil Nadu
UP
States
Gujarat
Maharashtra
Rajasthan
Tamil Nadu
UP
Top N
5
Data Source
total consumption
usage by region
Top N
Bottom N
2019 Monthwise consumption
2020 Monthwise consumption
total region consumption
Usage before during lockdown
Monthwise Top consumption
120 marks
24 rows by 5 columns
SUM(Usage): 798,678
22°C
Clear
Search
ENG
INTL
21:19
11-02-2026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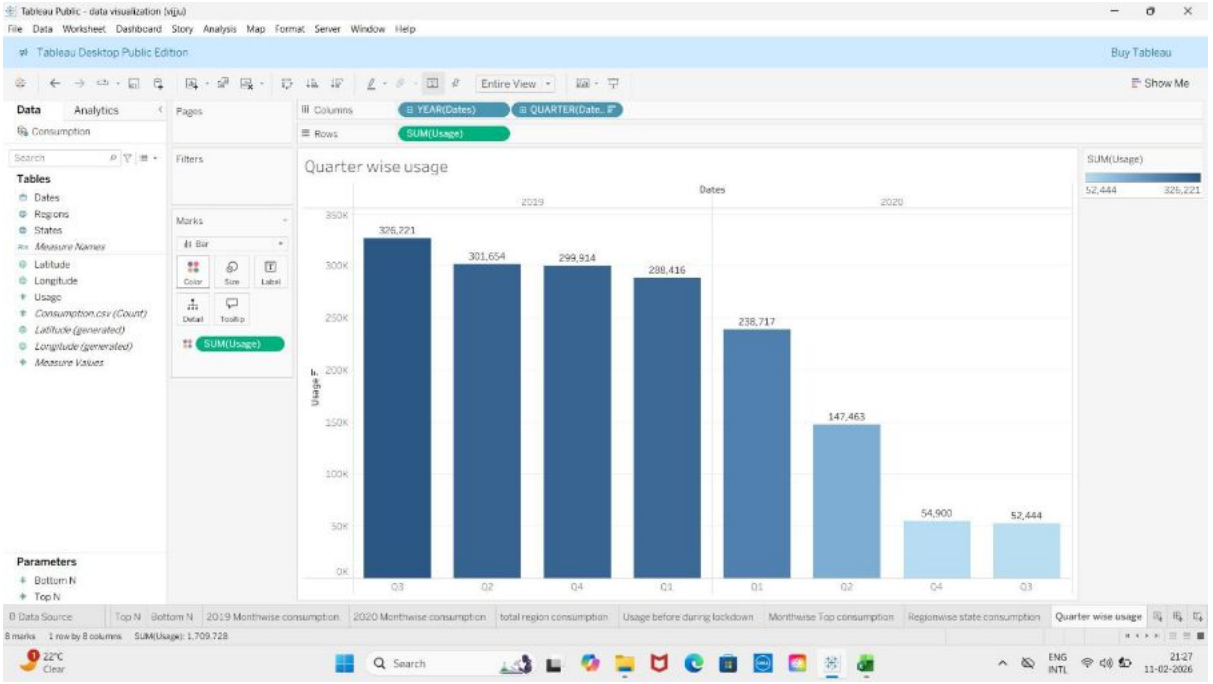
Automatic
Color
Size
Label
Detail
Tooltip
States
Columns
YEAR(Dates)
SUM(Usage)
Rows
Regions
States
Regionwise state consumption
Dates
2019
2020
Usage
States
Andhra Pradesh
Arunachal Pradesh
Assam
Bihar
Chandigarh
Chhattisgarh
Delhi
DNH
Goa
Gujarat
Haryana
HP
J&K
Jharkhand
Karnataka
Kerala
Maharashtra
Manipur
Meghalaya
Mizoram
MP
Nagaland
Odisha
Pondy
Punjab
Rajasthan
Sikkim
Tamil Nadu
Telangana
Tripura
UP
Uttarakhand
0 Data Source
usage by region
Top N
Bottom N
2019 Monthwise consumption
2020 Monthwise consumption
total region consumption
Usage before during lockdown
Monthwise Top consumption
Regionwise state consumption
66 marks
33 rows by 2 columns
SUM(Usage): 1,709,728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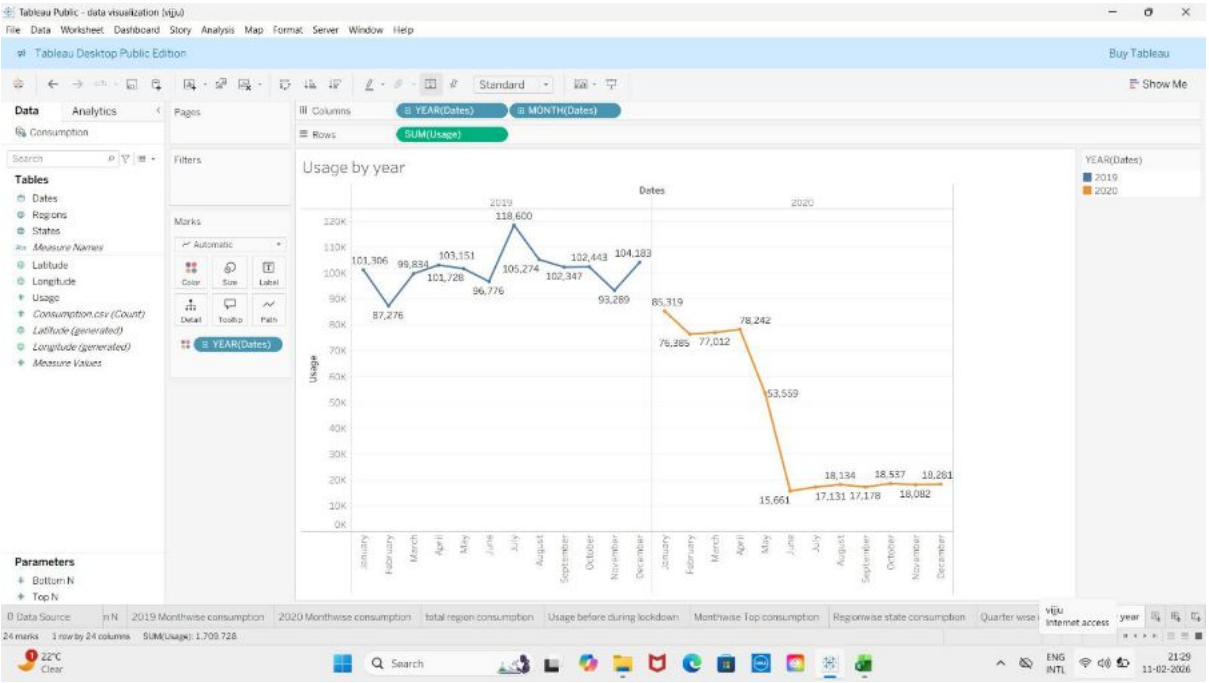
Longitude (generated)
Measure Values
Parameters
Bottom N
Top N
Pages
Filters
Marks
Bar
Color
Size
Label
Detail
Tooltip
SUM(Usage)
Columns
YEAR(Dates)
QUARTER(Date...
Rows
SUM(Usage)
Quarter wise usage
Dates
2019
2020
Usage
326,221
301,654
299,914
288,416
238,717
147,463
54,900
52,444
Q3
Q2
Q4
Q1
Q1
Q2
Q4
Q3
SUM(Usage)
52,444
326,221
0 Data Source
Top N
Bottom N
2019 Monthwise consumption
2020 Monthwise consumption
total region consumption
Usage before during lockdown
Monthwise Top consumption
Regionwise state consumption
Quarter wise usage
8 marks
1 row by 8 columns
SUM(Usage): 1,709,728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Marks
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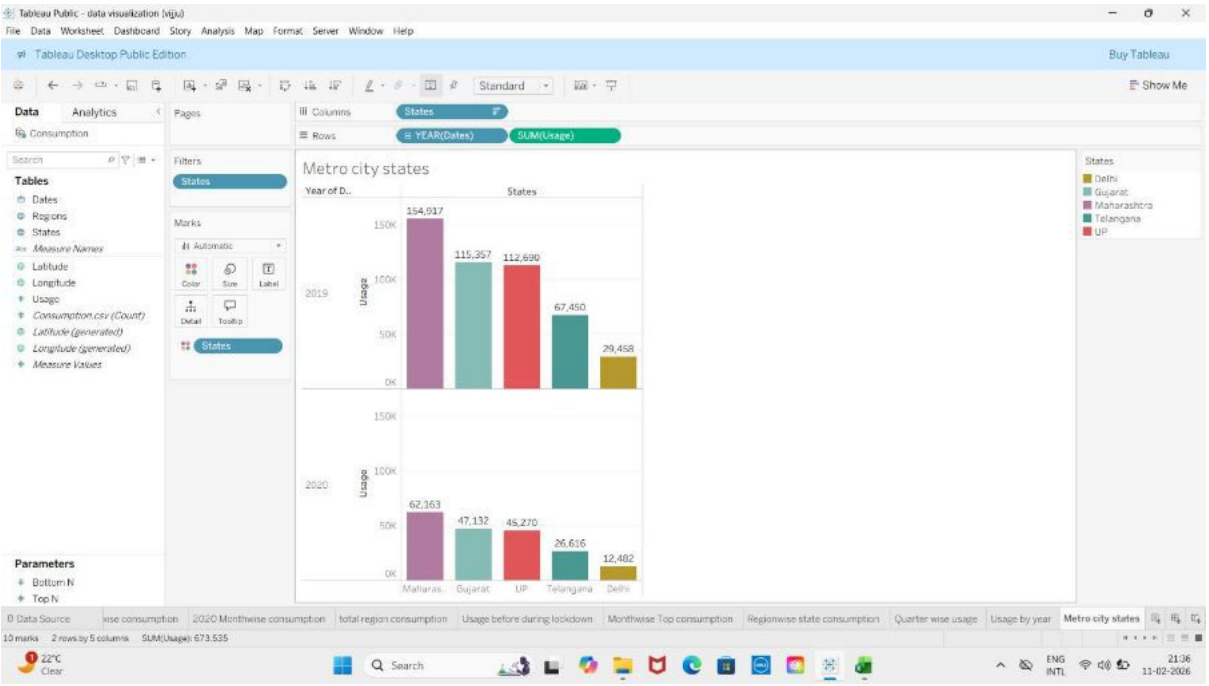
Size
Label
Detail
Tooltip
Path
YEAR(Dates)
Columns
YEAR(Dates)
MONTH(Dates)
Rows
SUM(Usage)
Usage by year
Dates
2019
2020
Usage
120K
110K
100K
90K
80K
70K
60K
50K
40K
30K
20K
10K
0K
101,306
87,276
99,834
101,728
103,151
96,776
118,600
105,274
102,347
102,443
93,289
104,183
85,319
76,285
77,012
78,242
53,559
15,661
17,131
17,178
18,134
18,537
18,082
18,281
January
February
March
April
May
June
July
August
September
October
November
December
YEAR(Dates)
2019
2020
0 Data Source
2019 Monthwise consumption
2020 Monthwise consumption
total region consumption
Usage before during lockdown
Monthwise Top consumption
Regionwise state consumption
Quarter wise
viju
Internet access
year
24 marks 1 row by 24 columns SUM(Usage): 1,709,728
22°C
Clear
Search
ENG
INTL
21:29
11-02-2026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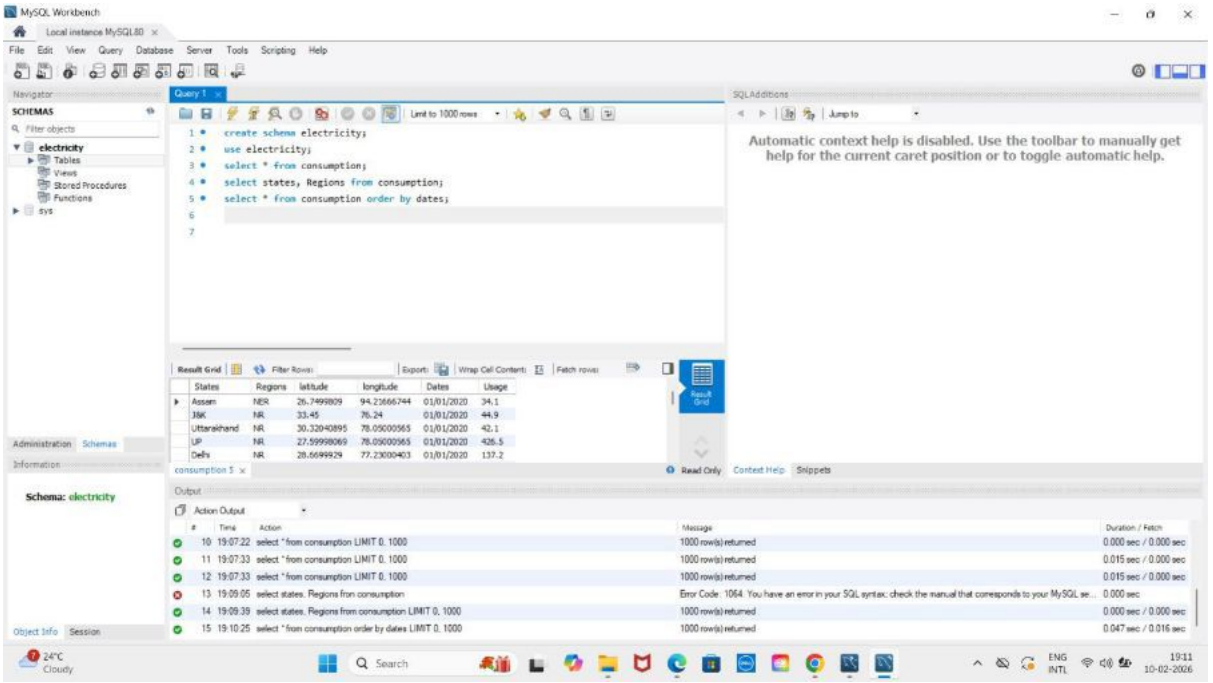
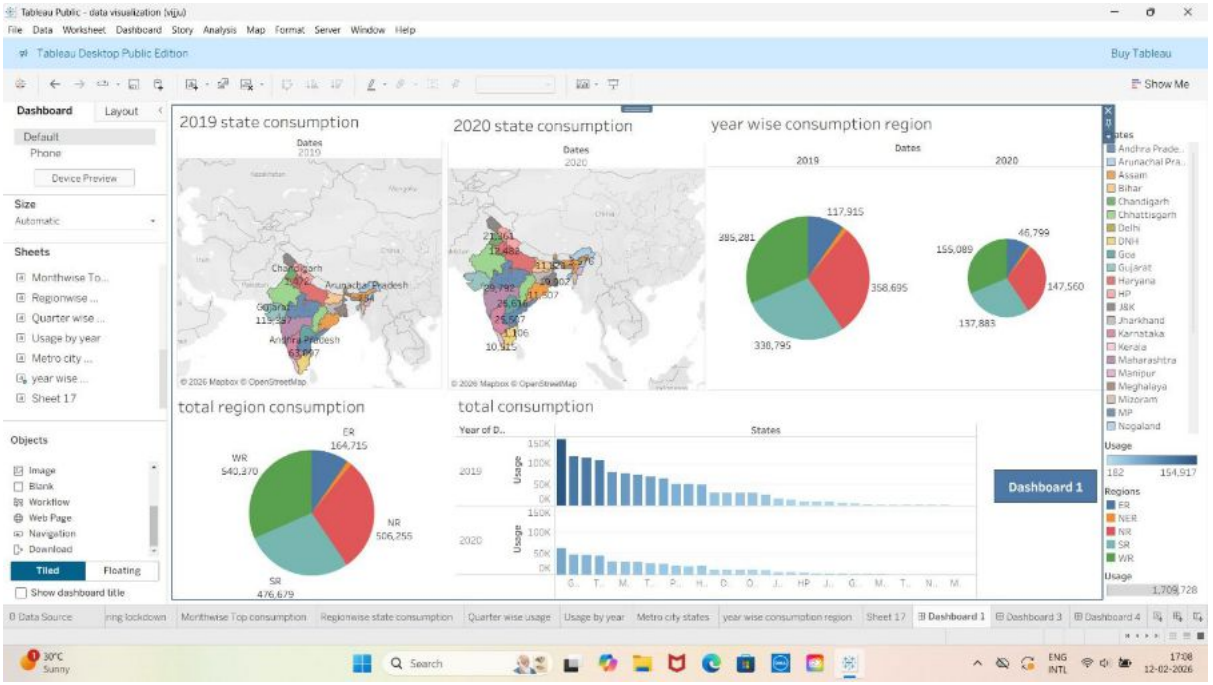
Size
Label
Detail
Tooltip
States
Columns
States
Rows
YEAR(Dates)
SUM(Usage)
Metro city states
Year of D..
States
2019
2020
Usage
150K
100K
50K
0K
154,917
115,357
112,690
67,450
29,458
62,163
47,132
45,270
26,616
12,482
Maharas..
Gujarat
UP
Telangana
Delhi
States
Delhi
Gujarat
Maharashtra
Telangana
UP
0 Data Source
2020 Monthwise consumption
total region consumption
Usage before during lockdown
Monthwise Top consumption
Regionwise state consumption
Quarter wise usage
Usage by year
Metro city states
10 marks 2 rows by 5 columns SUM(Usage): 673,535
22°C
Clear
Search
ENG
INTL
21:36
11-02-2026



Screenshots of Dashboard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Regionwise ...
Quarter wise ...
Usage by year
Metro city ...
year wise ...
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Workflow
Web Page
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Floating
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2019 state consumption
Dates
2019
2020 state consumption
Dates
2020
year wise consumption region
Dates
2019
2020
117,915
295,281
358,695
338,795
46,799
155,089
147,560
137,883
total region consumption
ER
164,715
WR
540,270
NR
506,255
SR
476,679
total consumption
Year of D..
States
2019
2020
Usage
Dashboard 1
Regions
ER
NER
NR
SR
WR
Usage
1,709,728
Data Source
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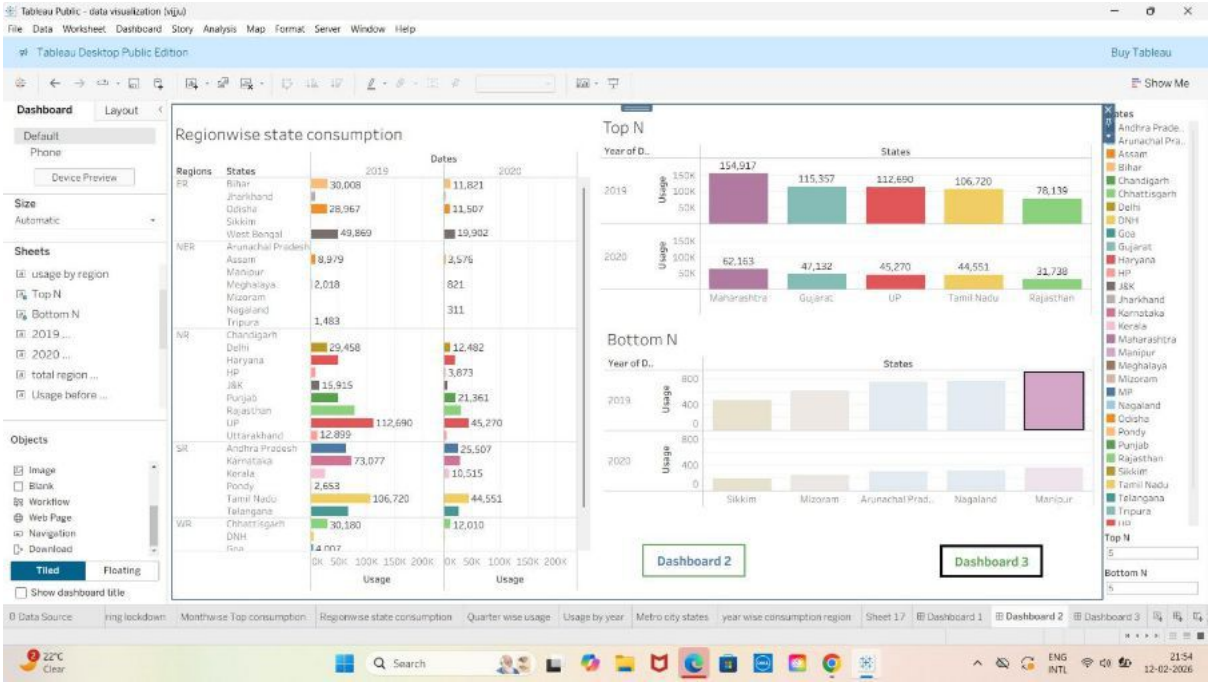
Regionwise state consumption
Quarter wise usage
Usage by year
Metro city states
year wise consumption region
Sheet 17
Dashboard 1
Dashboard 3
Dashboard 4
30°C
Sunny
Search
ENG
INTL
17:08
12-02-2026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Sheets
usage by region
Top N
Bottom N
2019 ...
2020 ...
total region ...
Usage before ...
Objects
Image
Blank
Workflow
Web Page
Navigation
Download
Tiled
Floating
Show dashboard title
Regionwise state consumption
Regions
States
Dates
2019
2020
ER
Bihar
Jharkhand
Odisha
Sikkim
West Bengal
30,008
28,967
49,869
11,821
11,507
19,902
NER
Arunachal Pradesh
Assam
Manipur
Meghalaya
Mizoram
Nagaland
Tripura
8,979
2,018
1,483
2,576
821
311
NR
Chandigarh
Delhi
Haryana
HP
J&K
Punjab
Rajasthan
UP
Uttarakhand
29,458
15,915
112,690
12,899
12,482
3,873
21,361
45,270
SR
Andhra Pradesh
Karnataka
Kerala
Pondy
Tamil Nadu
Telangana
73,077
2,653
106,720
25,507
10,515
44,551
WR
Chhattisgarh
DNH
30,180
12,010
Usage
Top N
Year of D..
States
2019
2020
154,917
115,357
112,690
106,720
78,139
62,163
47,132
45,270
44,551
31,738
Maharashtra
Gujarat
UP
Tamil Nadu
Rajasthan
Bottom N
Sikkim
Mizoram
Arunachal Prad..
Nagaland
Manipur
Dashboard 2
Dashboard 3
Top N
5
Bottom N
5
Data Source
Monthwise Top consumption
Regionwise state consumption
Quarter wise usage
Usage by year
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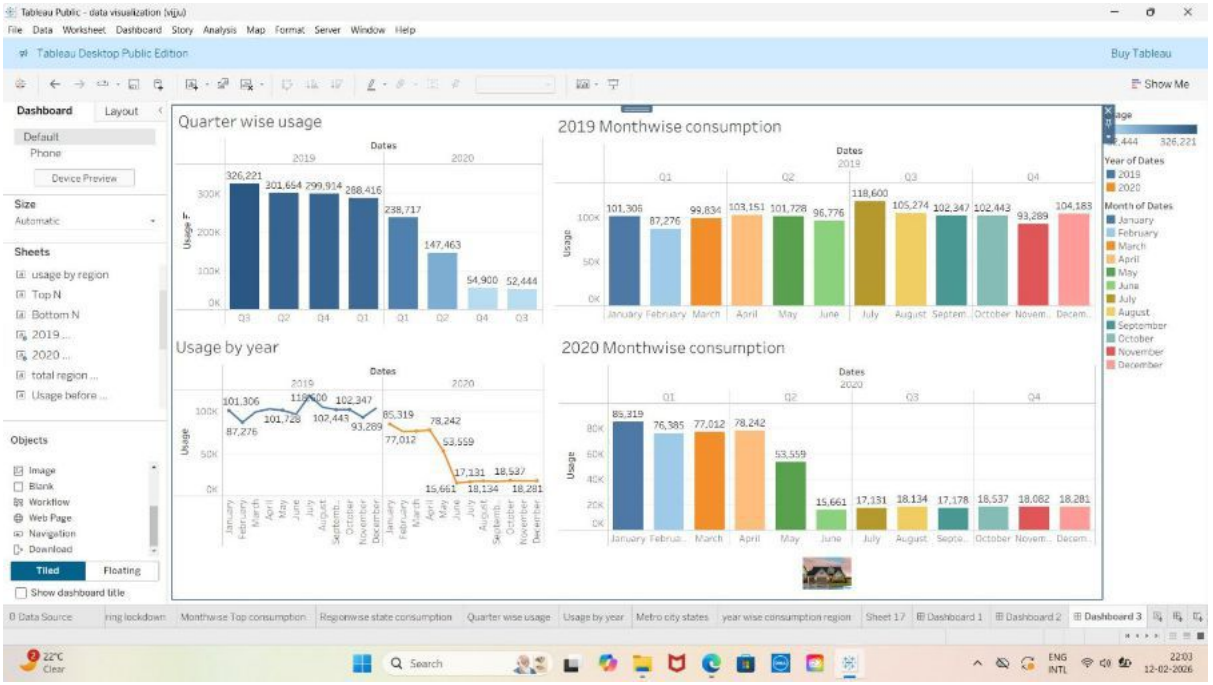
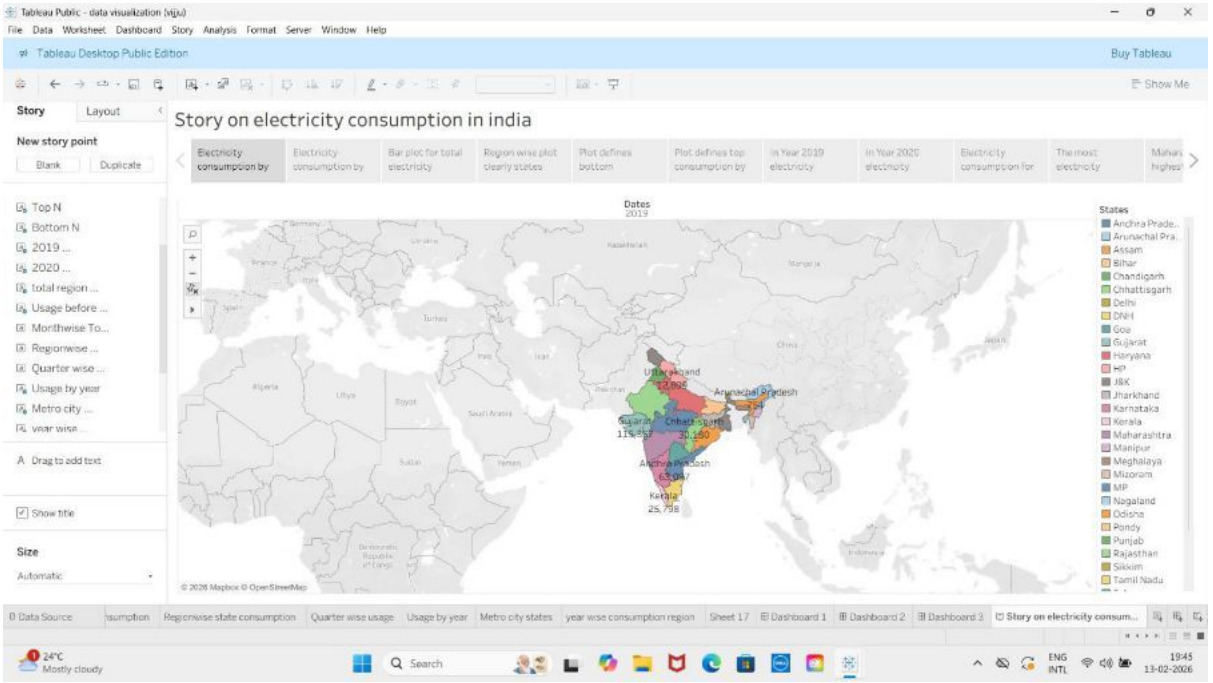
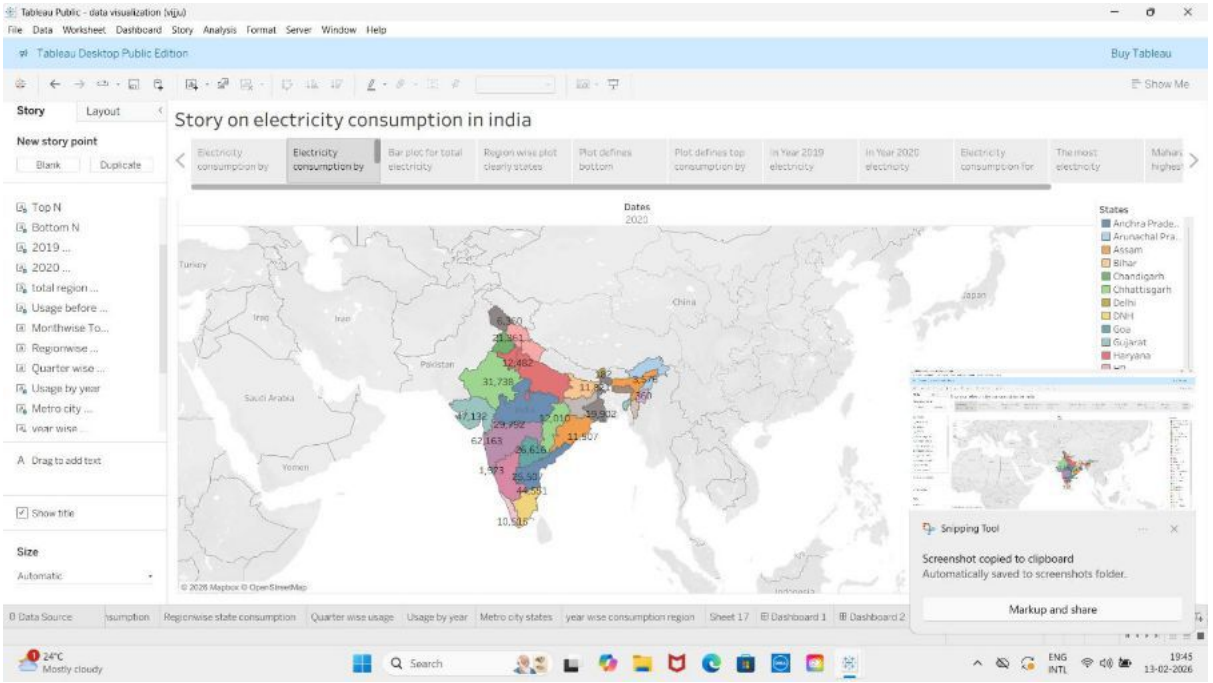
year wise consumption region
Sheet 17
Dashboard 1
Dashboard 2
Dashboard 3
22°C
Clear
Search
ENG
INTL
21:54
12-02-2026



Screenshots of story

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New story point
Blank
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Story on electricity consumption in india
Electricity consumption by
Electricity consumption by
Bar plot for total electricity
Region wise plot clearly states
Plot defines bottom
Plot defines top consumption by
In Year 2019 electricity
In Year 2020 electricity
Electricity consumption for
The most electricity
Top N
Bottom N
2019 ...
2020 ...
total region ...
Usage before ...
Monthwise To...
Regionwise ...
Quarter wise ...
Usage by year
Metro city ...
year wise ...
Drag to add text
Show title
Size
Automatic
Dates
2020
States
Anchra Prade..
Arunachal Pra..
Assam
Bihar
Chandigarh
Chhattisgarh
Delhi
DNH
Goa
Gujarat
Haryana
Turkey
Iraq
Iran
Pakistan
China
Japan
Saudi Arabia
Yemen
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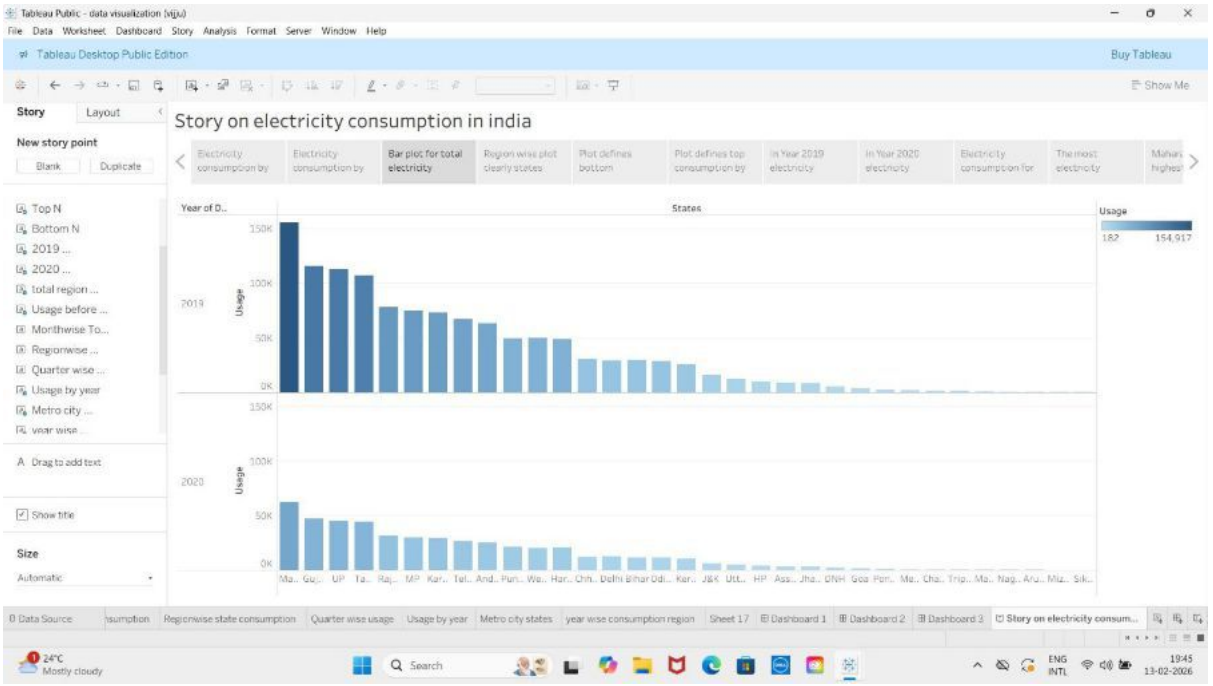
0 Data Source
Regionwise state consumption
Quarter wise usage
Usage by year
Metro city states
year wise consumption region
Sheet 17
Dashboard 1
Dashboard 2
24°C
Mostly cloudy
Search
ENG
INTL
19:45
13-02-2026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Bar plot for total electricity
Region wise plot clearly states
Plot defines bottom
Plot defines top consumption by
In Year 2019 electricity
In Year 2020 electricity
Electricity consumption for
The most electricity
Top N
Bottom N
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total region ...
Usage before ...
Monthwise To...
Regionwise ...
Quarter wise ...
Usage by year
Metro city ...
year wise ...
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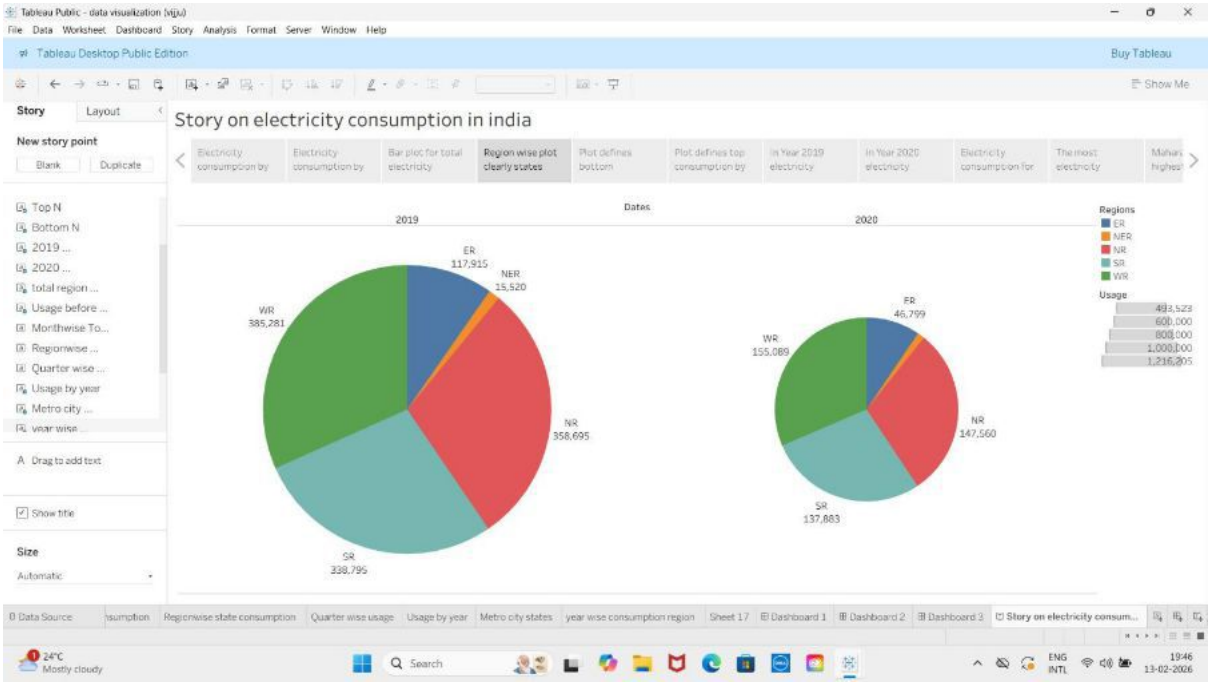
States
Usage
182
154,917
2019
2020
Usage
150K
100K
50K
0K
Ma.. Gu.. UP Ta.. Raj.. MP Kar.. Tel.. And.. Pun.. We.. Har.. Chh.. Delhi Bihar Odi.. Ker.. J&K Utt.. HP Ass.. Jha.. DNH Goa Pon.. Me.. Cha.. Tri.. Ma.. Nag.. Aru.. Mi.. Sik..
0 Data Source
Regionwise state consumption
Quarter wise usage
Usage by year
Metro city states
year wise consumption region
Sheet 17
Dashboard 1
Dashboard 2
Dashboard 2
Story on electricity consum...
24°C
Mostly cloudy
Search
ENG
INTL
19:45
13-02-2026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Region wise plot clearly states
Plot defines bottom
Plot defines top consumption by
In Year 2019 electricity
In Year 2020 electricity
Electricity consumption for
The most electricity
Top N
Bottom N
2019 ...
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total region ...
Usage before ...
Monthwise To...
Regionwise ...
Quarter wise ...
Usage by year
Metro city ...
year wise ...
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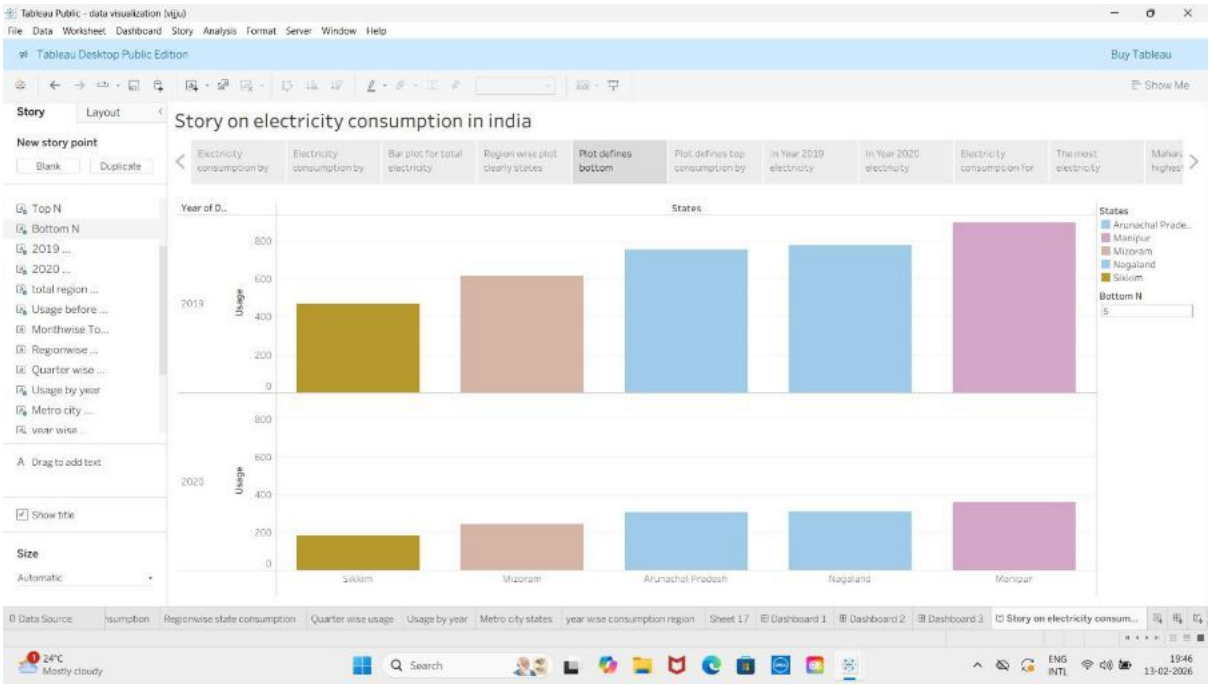
Size
Automatic
Dates
2019
2020
ER
117,915
NER
15,520
NR
358,695
SR
338,799
WR
385,281
ER
46,799
NR
147,560
SR
137,883
WR
155,089
Regions
ER
NER
NR
SR
WR
Usage
483,529
600,000
800,000
1,000,000
1,216,205
0 Data Source
Regionwise state consumption
Quarter wise usage
Usage by year
Metro city states
year wise consumption region
Sheet 17
Dashboard 1
Dashboard 2
Dashboard 2
Story on electricity consum...
24°C
Mostly cloudy
Search
ENG
INTL
19:46
13-02-2026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Story
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New story point
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Story on electricity consumption in india
Electricity consumption by
Electricity consumption by
Bar plot for total electricity
Region wise plot clearly states
Plot defines bottom
Plot defines top consumption by
In Year 2019 electricity
In Year 2020 electricity
Electricity consumption for
The most electricity
Top N
Bottom N
2019 ...
2020 ...
total region ...
Usage before ...
Monthwise To...
Regionwise ...
Quarter wise ...
Usage by year
Metro city ...
year wise ...
Drag to add text
Show title
Size
Automatic
Year of D..
States
2019
2020
Usage
800
600
400
200
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Sikkim
Mizoram
Arunachal Pradesh
Nagaland
Manipur
States
Arunachal Prade..
Manipur
Mizoram
Nagaland
Sikkim
Bottom N
5
0 Data Source
Regionwise state consumption
Quarter wise usage
Usage by year
Metro city states
year wise consumption region
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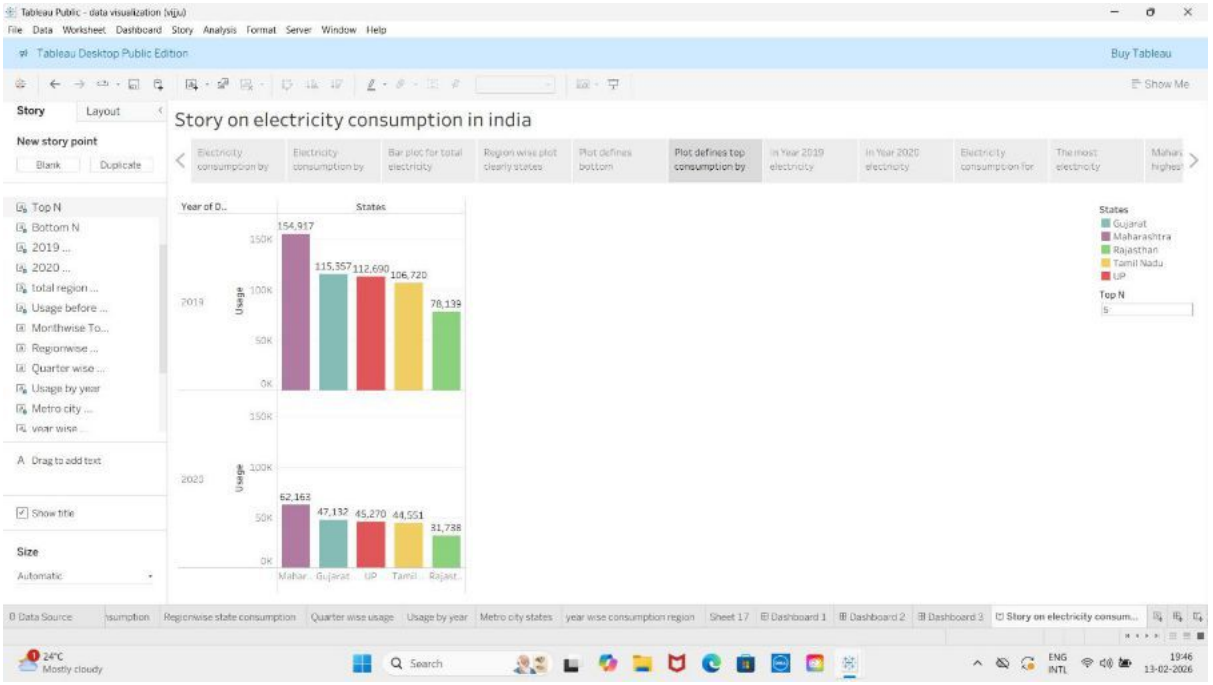
Dashboard 1
Dashboard 2
Dashboard 2
Story on electricity consum...
24°C
Mostly cloudy
Search
ENG
INTL
19:46
13-02-2026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Story on electricity consumption in india
Top N
Bottom N
2019 ...
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Usage before ...
Monthwise To...
Regionwise ...
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Metro city ...
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Drag to add text
Show title
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Usage
154,917
115,357
112,690
106,720
78,139
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45,270
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Mahar..
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UP
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Regionwise state consumption
Quarter wise usage
Usage by year
Metro city states
year wise consumption region
Sheet 17
Dashboard 1
Dashboard 2
Story on electricity consum...
24°C
Mostly cloudy
Search
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Tableau Public - data visualization (viju)
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Tableau Desktop Public Edition
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Show Me
Story
Layout
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Story on electricity consumption in india
Dates
2019
Q1
Q2
Q3
Q4
Usage
101,306
87,276
99,834
103,151
101,728
96,776
118,600
105,274
102,247
102,443
93,289
104,183
January
February
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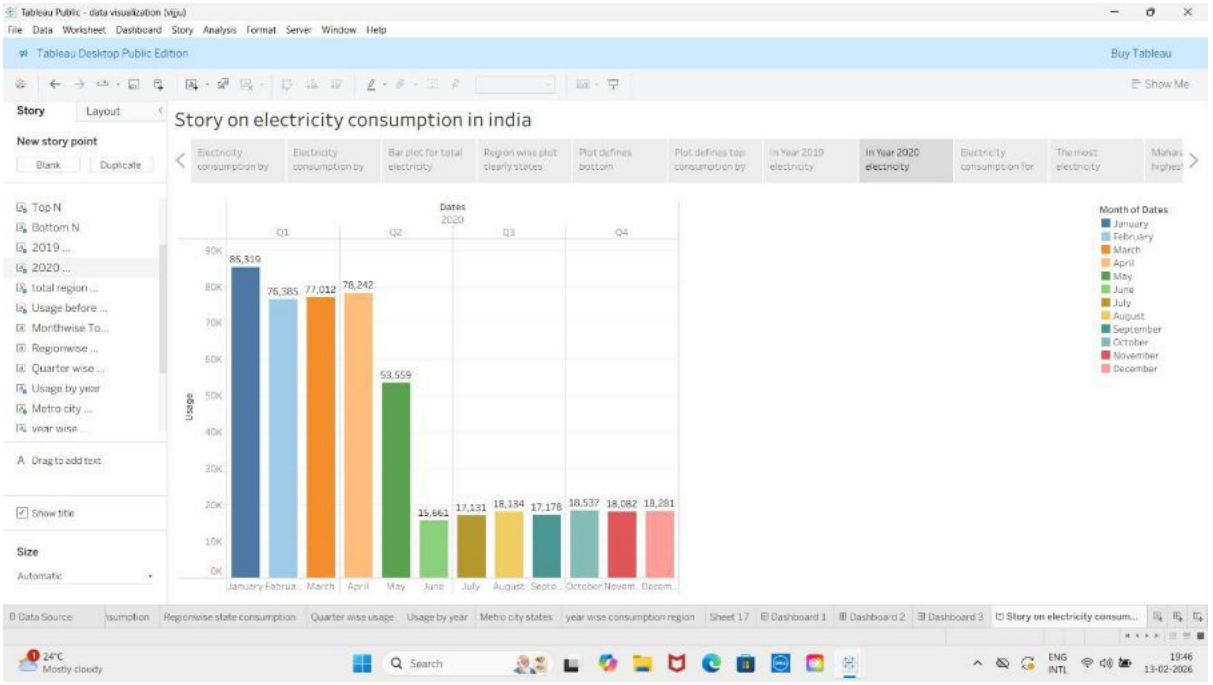
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year wise ...
Drag to add text
Show title
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Dates
2020
Q1
Q2
Q3
Q4
Usage
85,319
76,385
77,012
78,242
53,559
15,661
17,131
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Quarter wise usage
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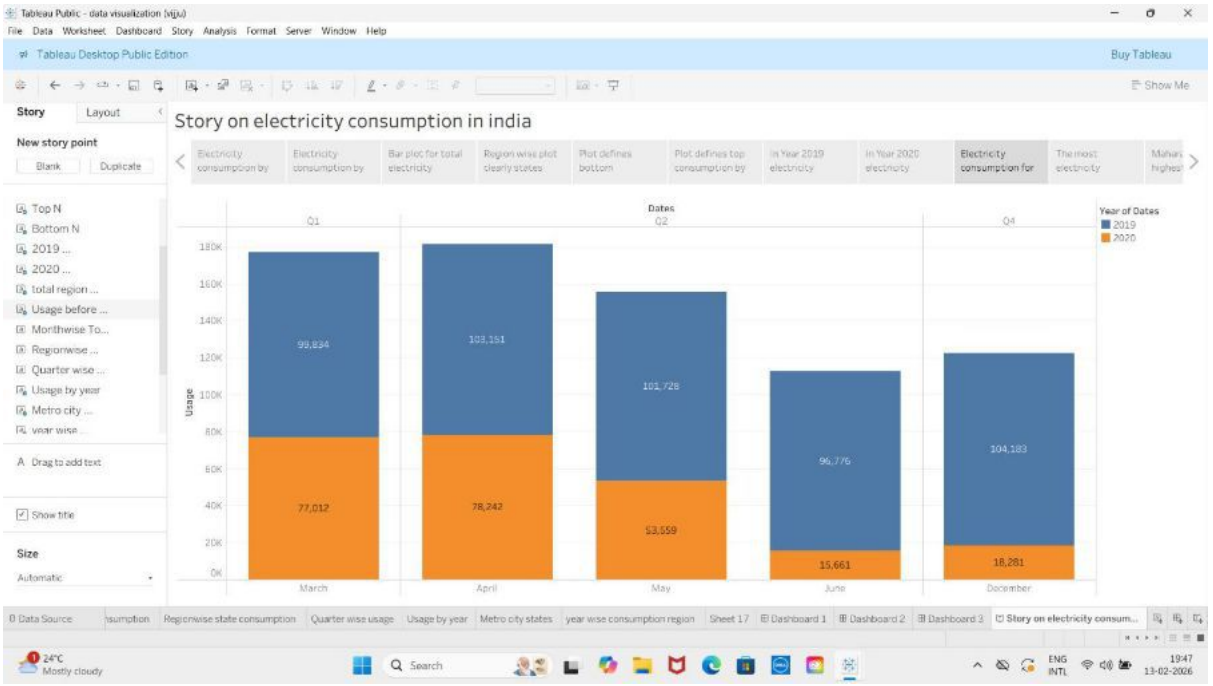
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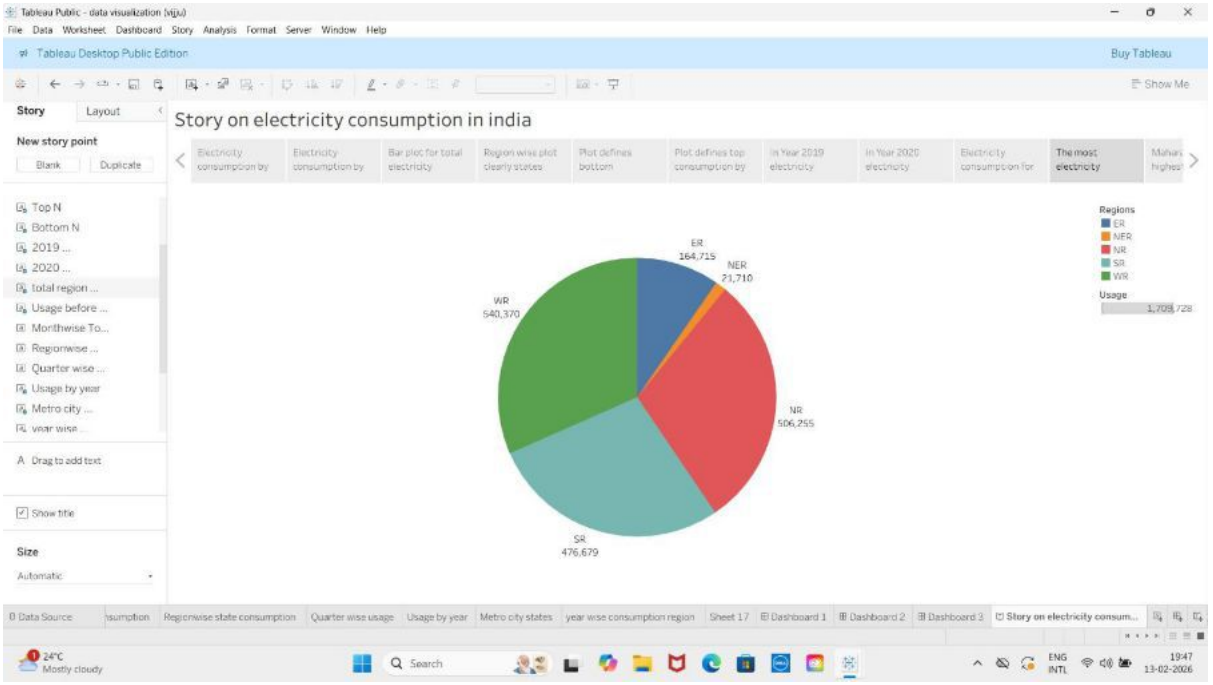
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Tableau Desktop Public Edition
Buy Tableau
Show Me
Story
Layout
New story point
Blank
Duplicate
Story on electricity consumption in india
Electricity consumption by
Electricity consumption by
Bar plot for total electricity
Region wise plot clearly states
Plot defines bottom
Plot defines top consumption by
In Year 2019 electricity
In Year 2020 electricity
Electricity consumption for
The most electricity
Top N
Bottom N
2019 ...
2020 ...
total region ...
Usage before ...
Monthwise To...
Regionwise ...
Quarter wise ...
Usage by year
Metro city ...
year wise ...
Drag to add text
Show title
Size
Automatic
ER
164,715
NER
21,710
NR
506,255
SR
476,679
WR
540,370
Regions
ER
NER
NR
SR
WR
Usage
1,709,728
Data Source
Regionwise state consumption
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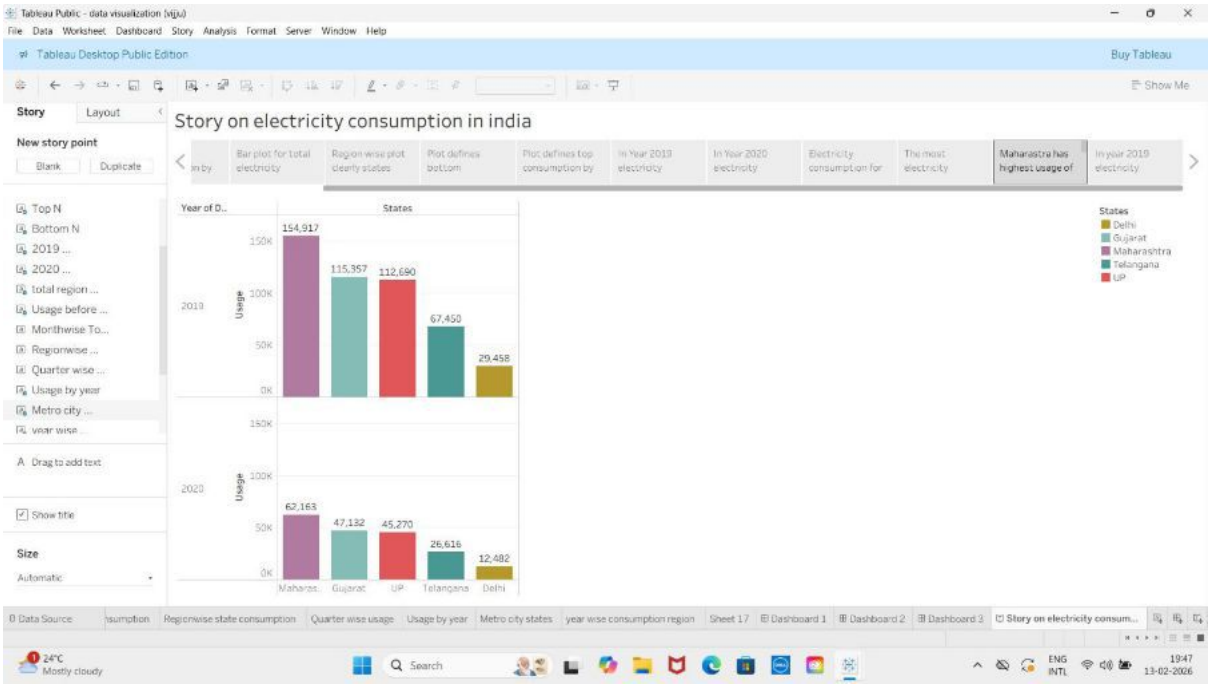
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Duplicate
Story on electricity consumption in india
Bar plot for total electricity
Region wise plot clearly states
Plot defines bottom
Plot defines top consumption by
In Year 2019 electricity
In Year 2020 electricity
Electricity consumption for
The most electricity
Maharastra has highest usage of
In year 2019 electricity
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2019 ...
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total region ...
Usage before ...
Monthwise To...
Regionwise ...
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Usage by year
Metro city ...
year wise ...
Drag to add text
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Size
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Year of D..
States
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2020
Usage
150K
100K
50K
0K
154,917
115,357
112,690
67,450
29,458
62,163
47,132
45,270
26,616
12,482
Maharas..
Gujarat
UP
Telangana
Delhi
States
Delhi
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Maharashtra
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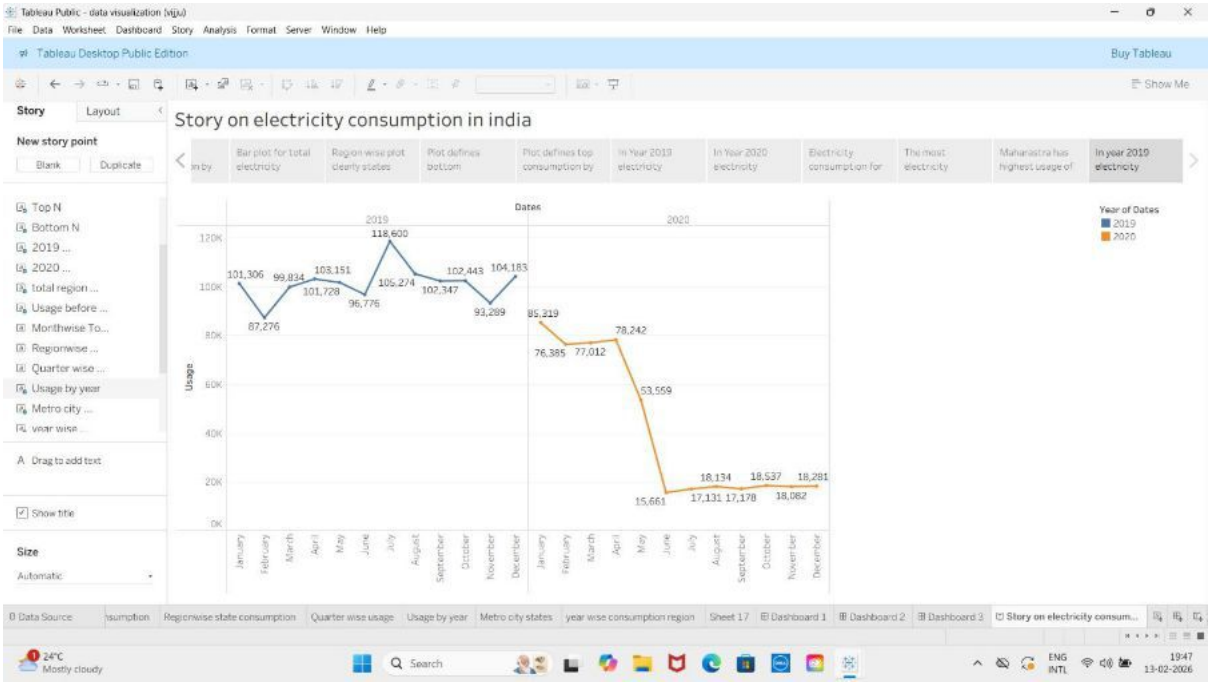

ADVANTAGES AND DISADVANTAGES

Advantages:

- **Data Visualization Clarity**

Using Tableau makes complex electricity data easy to understand through charts, graphs, and dashboards.

- **Trend Analysis**

Helps identify daily, monthly, and yearly electricity consumption patterns.

Useful for forecasting future energy needs.

- **Better Decision-Making**

Government, businesses, and households can make informed decisions about energy saving and cost control.

- **Interactive Dashboards**

Users can filter by region, time period, or sector (residential, industrial, etc.).

Makes analysis more flexible and user-friendly.

- **Energy Efficiency Awareness**

Highlights high consumption areas.

Encourages sustainable energy usage.

- **Cost Optimization**

Helps detect peak usage times, reducing electricity bills by planning usage efficiently.

Disadvantages:

- **Data Dependency**

Accuracy depends on the quality and completeness of collected electricity data.

- **Requires Technical Skills**

Knowledge of data cleaning, analysis, and Tableau tools is required.

- **Limited Predictive Capability**

Basic dashboards show past trends; advanced forecasting requires additional tools or models.

- **Data Privacy Concerns**

Electricity consumption data may reveal user behavior patterns.

- **Cost Factor**

A full version of Tableau can be expensive for small organizations.

- **Internet/System Requirement**

Requires good system performance and sometimes internet connectivity for sharing dashboards.

CONCLUSION

It provides valuable insights into electricity usage trends across different regions, sectors, and time periods. Using Tableau, interactive dashboards were created to visualize consumption patterns, identify peak usage periods, and analyze growth trends.

The analysis highlights how electricity demand changes over time and emphasizes the importance of efficient energy management and sustainable planning. By transforming raw data into meaningful visual insights, this project supports better decision-making for policymakers, utility providers, and consumers.

Overall, the study demonstrates how data visualization tools can help in understanding energy consumption behavior and contribute toward building a smarter, more energy-efficient future.

FUTURE SCOPE

- **Integration of Real-Time Data**

The project can be enhanced by integrating real-time electricity consumption data from smart meters and IoT devices. This will allow live dashboards in Tableau for instant monitoring and quicker decision-making.

- **Predictive Analytics Implementation**

Future work can include machine learning models to forecast electricity demand based on historical trends, weather conditions, and seasonal variations. This will help in better energy planning and load management.

- **Renewable Energy Analysis**

The dashboard can be expanded to analyze renewable energy usage (solar, wind, hydro) and compare it with conventional power consumption to support sustainable energy policies.

- **Cost Optimization Models**

Adding tariff-based analysis can help households and industries identify peak usage times and reduce electricity bills through optimized consumption strategies.

- **Regional & Sector-wise Expansion**

The project can be extended to compare electricity consumption across different states, cities, or sectors (residential, industrial, commercial) for deeper insights.

- **Carbon Emission Tracking**

Future versions can integrate carbon footprint calculations to evaluate environmental impact and promote green energy adoption.

- **Mobile & Web Deployment**

Publishing dashboards through Tableau Server or Tableau Public will enable wider accessibility for policymakers, researchers, and consumers.

Appendix: